

j6qthrmsx

May 3, 2024

```
[1]: # This Python 3 environment comes with many helpful analytics libraries
      ↳ installed
      # It is defined by the kaggle/python Docker image: https://github.com/kaggle/
      ↳ docker-python
      # For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/" directory
# For example, running this (by clicking run or pressing Shift+Enter) will list
↳ all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/) that
↳ gets preserved as output when you create a version using "Save & Run All"
# You can also write temporary files to /kaggle/temp/, but they won't be saved
↳ outside of the current session
```

```
/kaggle/input/publications/LEE.pdf
/kaggle/input/publications/Dawson.pdf
/kaggle/input/publications/Qiu.pdf
/kaggle/input/publications/1_Ramrez-Duque_.pdf
/kaggle/input/publications/Abbas_2020.pdf
/kaggle/input/publications/22_Ouss_ASD.pdf
/kaggle/input/publications/Asd_Cry_patterns.pdf
/kaggle/input/publications/zhao2020.pdf
/kaggle/input/publications/Abbas_2018.pdf
/kaggle/input/publications/carpenter2020 (1).pdf
/kaggle/input/publications/Young_Behavior.pdf
/kaggle/input/publications/Tariq2018.pdf
/kaggle/input/publications/Patten_Audio.pdf
```

Installing all the dependencies

```
[2]: !pip install -q unstructured
      !pip install -q sentence-transformers
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

keras-cv 0.8.2 requires keras-core, which is not installed.

keras-nlp 0.9.3 requires keras-core, which is not installed.

tensorflow-decision-forests 1.8.1 requires wurlitzer, which is not installed.

google-cloud-bigquery 2.34.4 requires packaging<22.0dev,>=14.3, but you have packaging 24.0 which is incompatible.

jupyterlab 4.1.6 requires jupyter-lsp>=2.0.0, but you have jupyter-lsp 1.5.1 which is incompatible.

jupyterlab-lsp 5.1.0 requires jupyter-lsp>=2.0.0, but you have jupyter-lsp 1.5.1 which is incompatible.

libpysal 4.9.2 requires shapely>=2.0.1, but you have shapely 1.8.5.post1 which is incompatible.

momepy 0.7.0 requires shapely>=2, but you have shapely 1.8.5.post1 which is incompatible.

osmnx 1.9.2 requires shapely>=2.0, but you have shapely 1.8.5.post1 which is incompatible.

spopt 0.6.0 requires shapely>=2.0.1, but you have shapely 1.8.5.post1 which is incompatible.

tensorflow 2.15.0 requires keras<2.16,>=2.15.0, but you have keras 3.2.1 which is incompatible.

ydata-profiling 4.6.4 requires numpy<1.26,>=1.16.0, but you have numpy 1.26.4 which is incompatible.

Importing all the dependencies

```
[3]: import warnings
      warnings.filterwarnings('ignore')
      import os
      import json
      from unstructured_client import UnstructuredClient
```

```

from unstructured_client.models import shared
from unstructured_client.models.errors import SDKError
from unstructured.staging.base import dict_to_elements, elements_to_json
from IPython.display import JSON

```

Setting up the user secrets and unstructured client

```

[4]: from kaggle_secrets import UserSecretsClient
user_secrets = UserSecretsClient()
secret_value_0 = user_secrets.get_secret("uns_api")
secret_value_1 = user_secrets.get_secret("uns_server_url")
s = UnstructuredClient(
    api_key_auth=secret_value_0,
    server_url=secret_value_1,
)

```

Getting elements from a single pdf using the Unstructured client

```

[5]: def get_elements_from_pdf(filename):
    with open(filename, "rb") as f:
        files=shared.Files(
            content=f.read(),
            file_name=filename,
        )

    req = shared.PartitionParameters(
        files=files,
        strategy='hi_res',
        pdf_infer_table_structure=True,
        languages=["eng"],
    )
    try:
        resp = s.general.partition(req)
    except SDKError as e:
        print(e)
    return resp

filename = "/kaggle/input/publications/1_Ramrez-Duque_.pdf"
response = get_elements_from_pdf(filename)
print(json.dumps(response.elements[:3], indent=2))

```

```

[
  {
    "type": "Header",
    "element_id": "124519724f8942a945cd489488e97294",
    "text": "Journal of Intelligent & Robotic Systems (2019) 96:267-281",
    "https://doi.org/10.1007/510846-018-00975-y",
    "metadata": {
      "filetype": "application/pdf",

```

```

        "languages": [
            "eng"
        ],
        "page_number": 1,
        "filename": "1_Ramrez-Duque_.pdf"
    }
},
{
    "type": "Title",
    "element_id": "0606ba94dc82b8248e73d33d5dce1c37",
    "text": "Robot-Assisted Autism Spectrum Disorder Diagnostic Based on
Artificial Reasoning",
    "metadata": {
        "filetype": "application/pdf",
        "languages": [
            "eng"
        ],
        "page_number": 1,
        "parent_id": "124519724f8942a945cd489488e97294",
        "filename": "1_Ramrez-Duque_.pdf"
    }
},
{
    "type": "Title",
    "element_id": "165abf064c3a95bbbee57e8aa4016140c",
    "text": "Andr\u00e9s A. Ramirez-Duque\u2019 @ . Anselmo Frizera-Neto' -
Teodiano Freire Bastos\u2019",
    "metadata": {
        "filetype": "application/pdf",
        "languages": [
            "eng"
        ],
        "page_number": 1,
        "parent_id": "124519724f8942a945cd489488e97294",
        "filename": "1_Ramrez-Duque_.pdf"
    }
}
}
]

```

```
[6]: JSON(json.dumps(response.elements, indent=2))
```

```
[6]: <IPython.core.display.JSON object>
```

```
[7]: for element in response.elements:
    if element['type'] == "Title" and element['text'] == "References":
        print('element id: ', element['element_id'])
        print('parent id: ', element['metadata']['parent_id'])
```

element id: 69824d3b0e70ca6aaa0da1613b65fd91
parent id: efd96aedef377e20afd95285a7c751a86

```
[8]: #Extracting text out of this pdf without the references, headers and footers
def cleaned_text_without_references(response):
    cleaned_elements=[]
    for element in response.elements:
        if element['text'] == 'REFERENCES' or element['text']=='References':
            break
        elif element['type'] != "Header" and element['type'] != "Footer":
            cleaned_elements.append({'element_id': element['element_id'],
                                     'page_number': element['metadata']['page_number'],
                                     'text': element['text']})
    return cleaned_elements
cleaned_text = cleaned_text_without_references(response)
cleaned_text[:5]
```

```
[8]: [{'element_id': '0606ba94dc82b8248e73d33d5dce1c37',
      'page_number': 1,
      'text': 'Robot-Assisted Autism Spectrum Disorder Diagnostic Based on
Artificial Reasoning'},
      {'element_id': '165abf064c3a95bbee57e8aa4016140c',
      'page_number': 1,
      'text': "Andrés A. Ramirez-Duque' @ . Anselmo Frizera-Neto' - Teodiano Freire
Bastos'"},
      {'element_id': 'b2181c6eefd507a2a68fcd65e3af62e0',
      'page_number': 1,
      'text': '1.) Check for updates'},
      {'element_id': 'dbc91cf0825c7c18192c20fc0182d322',
      'page_number': 1,
      'text': 'Received: 25 April 2018 / Accepted: 20 December 2018 / Published
online: 29 March 2019 © Springer Nature B.V. 2019'},
      {'element_id': 'd21b4a64a2d8656a0fdf7ab2e89a4916',
      'page_number': 1,
      'text': 'Abstract'}]
```

```
[12]: #grouping all the text on same page
def group_by_page(cleaned_text):
    grouped_texts=[]
    #finding the max page number
    max_page_number = 0
    for item in cleaned_text:
        page_number = item['page_number']
        if page_number > max_page_number:
            max_page_number = page_number
    #grouping all the text on same page
```

```

for i in range(max_page_number):
    item_text = ''
    item_page_number = i+1
    for item in cleaned_text:
        if item['page_number'] == i+1:
            item_text = item_text + item['text']
    grouped_texts.append({'page_number': item_page_number,
                          'text': item_text})

return grouped_texts

grouped_text = group_by_page(cleaned_text)
grouped_text[:2]

```

[12]: [{'page_number': 1, 'text': "Robot-Assisted Autism Spectrum Disorder Diagnostic Based on Artificial ReasoningAndrés A. Ramirez-Duque' @ . Anselmo Frizera-Neto' - Teodiano Freire Bastos'l.) Check for updatesReceived: 25 April 2018 / Accepted: 20 December 2018 / Published online: 29 March 2019 © Springer Nature B.V. 2019AbstractAutism spectrum disorder (ASD) is a neurodevelopmental disorder that affects people from birth, whose symptoms are found in the early developmental period. The ASD diagnosis is usually performed through several sessions of behavioral observation, exhaustive screening, and manual coding behavior. The early detection of ASD signs in naturalistic behavioral observation may be improved through Child-Robot Interaction (CRI) and technological-based tools for automated behavior assessment. Robot-assisted tools using CRI theories have been of interest in intervention for children with Autism Spectrum Disorder (CwASD), elucidating faster and more significant gains from the diagnosis and therapeutic intervention when compared to classical methods. Additionally, using computer vision to analyze child's behaviors and automated video coding to summarize the responses would help clinicians to reduce the delay of ASD diagnosis. In this article, a CRI to enhance the traditional tools for ASD diagnosis is proposed. The system relies on computer vision and an unstructured and scalable network of RGBD sensors built upon Robot Operating System (ROS) and machine learning algorithms for automated face analysis. Also, a proof of concept is presented, with participation of three typically developing (TD) children and three children in risk of suffering from ASD.Keywords Child-Robot interaction - Autism spectrum disorder - Convolutional neural network - Robot reasoning model Statistical shape modeling1 IntroductionResearch in Child-Robot Interaction (CRI) aims to provide the necessary conditions for the interaction between a child and a robotic device taking into account some fundamental features, such as child's neurophysical and physical condition, and the child's mental health [1]. That is how Robot-Assisted Therapies (RAT) using CRI theories have been of interest as an intervention for CwASD, elucidating faster and more significant gains from the therapeutic intervention when compared to traditional therapies [2-4].ASD is a neurodevelopmental disorder that affects people from birth, and its symptoms are found in the early1 Universidade Federal do Espírito Santo., Av. Fernando Ferrari, 514 (29075-910), Vitoria, Brazildevelopmental period.

Individuals suffering from ASD exhibit persistent deficits in social communication, social interaction and repetitive patterns of behavior, interests, or activities [5]. Some of the ASD signs may be observed before the age of 10 months, although a reliable diagnosis can only be performed at 18 months of age, according to [6], or 24 months according to [7]. The use of computer vision to analyze the child's behaviors, and automated video coding to summarize the interventions, can help the clinicians to reduce the delay of ASD diagnosis, providing the CwASD with access to early therapeutic interventions. In addition, CRI-based intervention can transform traditional diagnosis methods through a robotic device to systematically elicit child's behaviors that exhibit ASD signs [8]. Some of the first systems developed to assist ASD therapists and make diagnosis based on robotic devices have primarily been open loop and remotely operated systems. However, these approaches are unable to perform autonomous feedback to enhance the interaction [9-11].

@ SpringerContent courtesy of Springer Nature, terms of use apply. Rights reserved."},
{'page_number': 2,

'text': 'Nevertheless, different systems are able to modify the behavior of the robot according to environmental interactions and the child's response, using a closed-loop and artificial cognition approaches [12--16]. These systems have been hypothesized to offer technological mechanisms for supporting more flexible and potentially more naturalistic interaction [17]. In fact, literature reports that automatic robot's social behaviors modulation according to specific scenarios has a strong effect on child's social behavior [12]. However, despite the increase of positive evidence, this technology has rarely been applied to specific ASD diagnosis. This work aims to present a robot-assisted framework using an artificial reasoning module to assist clinicians with the ASD diagnostic process. The framework is composed of a responsive robotic platform, a flexible and scalable vision sensor network, and an automated face analysis algorithm based on machine learning models. In this research we take advantage of some neural models available as open source projects to build a completely new pipeline algorithm for global recognition and tracking of child's face among many faces present in a typical unstructured clinical intervention, in order to estimate the child's visual focus of attention along the time. The proposed system can be used in different behavioral analysis scenarios typical of an ASD diagnostic process. In order to illustrate the feasibility of the proposed system, in this paper an experimental trial to assess joint-attention behavior is presented employing an in-clinic setup (unstructured environment). The main contributions of this paper are: (i) the development of a new artificial reasoning module upon a flexible and scalable ROS-based vision system using state-of-the-art machine learning neural models; (ii) the proposal and implementation of a supervised CRI (child-robot interaction) based on an open source social robotic platform to enhance the traditional tools for ASD diagnosis using an in-clinic setup protocol. For the best of our knowledge, there are no open source projects available for face analysis based on a multi-camera approach using ROS with the characteristics described in our research.

2 Related Work

Recent researches have shown the acceptance and efficiency of technologies used as auxiliary tools for therapy

and teaching of individuals with ASD [18-21]. Such technologies may also be useful for people surrounding ASD individuals (therapists, caregivers, family members). For example, the use of artificial vision systems to measure and analyze the child's behavior can lead to alternative screening and monitoring tools that help the clinicians to get feedback from the effectiveness of the intervention [22]. Additionally, social robots have great potential for aid in the diagnosis and therapy of children with ASD [18, 23]. A higher degree of control, prediction and simplicity may be achieved in interactions with robots, impacting directly on frustration and reducing the anxiety of these individuals [24]. Respect to the use of computer vision techniques, previous studies already analyzed child's behaviors, such as visual attention, eye gaze, eye contact, smile events, and visual exploration using cameras and eye trackers [25, 26] and RGBd cameras [27, 28]. These studies have shown the potential of vision systems in improving the behavioral coding in ASD therapies. However, these studies did not implement techniques of CRI to enhance the intervention. On the other hand, studies about how CwASD respond to a robot mediator compared to human mediator have been reported, such as intervention scenarios with imitation games [29, 30], telling stories [9] and free play tasks [12, 31]. These works used features, such as proxemics, body gestures, visual contact and eye gaze as behavioral descriptors, whereas the behavior analysis was estimated using manual video coding. Researchers of Vanderbilt University published a series of research showing an experimental protocol to assess joint attention (JA) tasks defined as the capacity for coordinated orientation of two people toward an object or event [6]. The protocol consisted of directing the attention of the child towards objects located in the room through adaptive prompts [32]. Bekele et al. inferred the participant's eye gaze by the head pose, which was calculated in real-time by an IR camera array [17]. In their last works, Zheng et al. and Warren et al. used a commercial eye tracker to estimate the children's eye gaze around the robot and manual behavioral coding for global evaluation [10, 33]. However, eye tracker devices require pre-calibration and may limit the movement of the individual. The results of these works showed that the robot attracted children's attention and that CwASD reached all JA task. Nevertheless, developing JA tasks is more difficult with a robot than with humans [10]. Anzalone et al. developed a CRI scenario using the NAO robot to perform JA tasks, in which the authors used an RGBD camera to estimate only body and head movements. The results showed that JA performance of children with ASD was similar to the performance of TD children when interacting with the human mediator, however, with a robot mediator, the children with ASD presented a lower performance than the TD children, i.e., the children with ASD needed more social cues to finalize the task [34]. Chevalier et al. analyzed in their study, some features, such as proprioceptive and visual integration in CwASD, using an RGBD sensor to record the interventions sessions and manual behavior coding to analyze the participants' performance [35]. In none of the previous works, a closed-loop subsystem was

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Extracting text out of each pdf without the references, headers and footers

```
[13]: pdf_and_text=[]
for filename in os.listdir("/kaggle/input/publications"):
    #check if the file is a pdf
    if filename.endswith('.pdf'):
        # Construct the full path to the PDF file
        pdf_path = os.path.join("/kaggle/input/publications", filename)
        pdf_response = get_elements_from_pdf(pdf_path)
        pdf_cleaned_text = cleaned_text_without_references(pdf_response)
        pdf_grouped_text = group_by_page(pdf_cleaned_text)
        pdf_and_text.append({'filename': filename.replace("/kaggle/input/
↪publications/", ""),
                            'text': pdf_grouped_text})

pdf_and_text[0]
```

```
[13]: {'filename': 'LEE.pdf',
       'text': [{'page_number': 1,
                  'text': '. sensorsbyLetterLetter Deep-Learning-Based Detection of Infants
with Autism Spectrum Disorder Using Auto-Encoder Feature RepresentationJung Hyuk
Lee 1, Geon Woo Lee 1, Guiyoung Bong 2, Hee Jeong Yoo 2,3 and Hong Kook Kim
1,*1School of Electrical Engineering and Computer Science, Gwangju Institute of
Science and Technology, Gwangju 61005, Korea; ljh0412@gist.ac.kr (J.H.L.);
geonwoo0801@gist.ac.kr (G.W.L.) 2 Department of Psychiatry, Seoul National
University Bundang Hospital, Seongnam-si,Gyeonggi-do 13620, Korea;
20409@snubh.org (G.B.); hjyoo@snu.ac.kr (H.J.Y.)3 Department of Psychiatry,
College of Medicine, Seoul National University, Seoul 03980, Korea *
Correspondence: hongkook@gist.ac.krReceived: 29 October 2020; Accepted: 24
November 2020; Published: 26 November 2020check for v updatesAbstract: Autism
spectrum disorder (ASD) is a developmental disorder with a life-span disability.
While diagnostic instruments have been developed and qualified based on the
accuracy of the discrimination of children with ASD from typical development
(TD) children, the stability of such procedures can be disrupted by limitations
pertaining to time expenses and the subjectivity of clinicians. Consequently,
automated diagnostic methods have been developed for acquiring objective
measures of autism, and in various elds of research, vocal characteristics have
not only been reported as distinctive characteristics by clinicians, but have
also shown promising performance in several studies utilizing deep learning
models based on the automated discrimination of children with ASD from children
with TD. However, difficulties still exist in terms of the characteristics of the
data, the complexity of the analysis, and the lack of arranged data caused by
the low accessibility for diagnosis and the need to secure anonymity. In order
to address these issues, we introduce a pre-trained feature extraction auto-
encoder model and a joint optimization scheme, which can achieve robustness for
widely distributed and unreined data using a deep-learning-based method for the
detection of autism that utilizes various models. By adopting this auto-encoder-
based feature extraction and joint optimization in the extended version of the
Geneva minimalistic acoustic parameter set (eGeMAPS) speech feature data set, we
acquire improved performance in the detection of ASD in infants compared to the
```

raw data set. Keywords: auto-encoder; bidirectional long short-term memory (BLSTM); joint optimization; acoustic feature extraction; autism spectrum disorder.

1. Introduction Autism spectrum disorder (ASD) is a developmental disorder with a high probability of causing difficulties in social interactions with other people [1]. According to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), ASD involves several characteristics such as being con ned to speci c interests or behaviors, delayed linguistic development, and poor functionality in terms of communicating or functioning in social situations [2]. As there is wide variation in terms of the types and severities of ASD based on its characteristics, the disorder is referred to as a spectrum [1]. Not only does ASD have the characteristics of a developmental disorder with a life-span disability, but its prevalence is also increasing—from 1 in 150 children in 2000 to 1 in 54 children in 2016 [3]. As diverse evidence has been obtained from previous research showing that the chance of improvement in the

Sensors 2020, 20, 6762;
doi:10.3390/s20236762www.mdpi.com/journal/sensors'},

```
{'page_number': 2,
```

'text': 'social abilities of people with ASD increases when an earlier clinical intervention is performed [4], the early detection of ASD characteristics has become a key point of current ASD research. Various instruments for discriminating ASD have been developed, and the commonly accepted gold standard schemes are behavioral assessments, which are time-consuming procedures and require multidisciplinary teams (MDTs). However, most behavioral assessments suffer in terms of the stability of their ASD diagnosis as a result of the issues of accessibility or subjectivity and interpretive bias between professions [5]. Therefore, several attempts to develop objective and precise diagnostic methods have been made in multiple elds, such as genetic determination [6], principle analysis of brain images [7], and physiological approaches [8]. One prominent area of behavioral observations is that of infants' vocal characteristics. Children with ASD are known to have abnormalities in their prosody resulting from deficits in their ability to recognize the inherent mental conditions of others [9], and their atypical vocalizations are known to be monotonous or exaggerated, which can be revealed using various acoustic characteristics, followed by engineering approaches for the discrimination of ASD or typical development (TD) in children based on the vocal and acoustic features. For example, in [10], the researchers estimated deficits in the vocalization of children with ASD at an average age of 18 months, such as "at" intonation, atypical pitch, or control of volume based on the variability of pitch and the long-term average spectrum (LTAS) using fast Fourier transform, where significant differences were observed in the spectral components at low-band frequencies, as well as spectral peaks and larger pitch ranges and standard deviations. The development of linguistic abilities is also considered to be a distinguishable feature of delayed development in children with ASD. Earlier vocal patterns at age 6–18 months were proven to be differentiable in a study [11] that aimed to confirm the hypothetical vocal patterns and social quality of vocal behavior in order to differentiate between ASD and TD cohorts in groups of children aged 0–6, 6–12, and 12–18 months in terms of categorized speech

patterns consisting of vocalization, long reduplicated babbling, two-syllable babbling, and rst words. Evidence of abnormalities in children with ASD were shown, in these cases, as a significant decrease in vocalization and rst word rate, while the difference in babbling ability between children with ASD and TD was negligible. Given the development and improvement of machine learning algorithms, as the achievement in the performance of state-of-the-art classification and discrimination tasks [12], recent attempts to develop automated classification methods based on machine learning techniques have been based on the distinctiveness of vocal characteristics, and have been shown to be promising alternatives to the conventional methods in many publications [13]. For examples of machine learning classification, the researchers of [14] employed various acoustic-prosodic features, including fundamental frequency, formant frequencies, harmonics, and root mean square signal energy. In their research, support vector machines (SVMs) and probabilistic neural networks (PNNs) were adopted as classifiers, which showed excellent accuracy in discriminating children with ASD from children with TD. Meanwhile, the authors of [15] employed more recent deep learning techniques, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) with spectral features from short-time Fourier transform (STFT) and constant Q transform (CQT), to classify children diagnosed using the autism diagnostic observation schedule (ADOS), also showing promising results in multiple outcomes from SVMs, RNNs, and a combination of CNN and RNN classifiers. A generalized acoustic feature set, an extended version of the Geneva minimalistic acoustic parameter set (eGeMAPS) [16], and the bidirectional long short-term memory (BLSTM) model were adopted to differentiate between children with ASD and children with TD in [17], showing that 75% of the subjects' utterances were correctly classified with the simple application of a deep learning model and feature sets. While the quality of previous research based on various acoustic features has proven the effectiveness of acoustic features and classification algorithms for the detection of abnormalities in children's voices in ASD group compared to those of TD group, the complexity and relationship being inherent between the features will remain uncertain until a large amount of data can be accumulated. Furthermore, a limitation still remains in terms of the problems regarding data collection, since there are

{'page_number': 3,

'text': 'difficulties pertaining to the need to secure the anonymity of infant subjects, as well as the unintended ignorance of parents at earlier stages of their infant's development. The data of infants are, accordingly, dispersed by gender, age, and number of vocalizations, or consist of comparably small volumes of audio engineering data in general. These problems were typically overlooked by previous research with controlled and small amounts of data. In order to provide suggestions for a method to overcome the abovementioned restrictions, we focus on examining the feasibility of neural networks as a feature extractor, employing an auto-encoder (AE), which can modify acoustic features into lowered and separable feature dimensions [18]. We construct a simple six-layered stacked AE that contains an input layer, three fully connected (FC) layers, an output layer, and one auxiliary output layer, which has categorical targets for ASD and TD for the optimization of the latent feature space of the AE. We train the AE

and deep learning models and compare the results for each model based on SVMs and vanilla BLSTM, while adopting the same model parameters from the method suggested in [17]. The remainder of this paper is organized as follows. Section 2 describes the specifications of the participants' data, data processing, feature extraction, statistical analysis, and experimental setup. Section 3 presents the performance evaluations for each algorithm of the SVMs and vanilla BLSTM. Lastly, Section 4 concludes the paper.

2. Proposed Method This study was based on the audio data from video recordings of ASD diagnoses, which were collected from 2016 to 2018 at Seoul National University Bundang Hospital (SNUBH). We received approval from the Institutional Review Board (IRB) at SNUBH to use fully anonymized data for retrospective analysis (IRB no: B-1909/567-110) from existing research (IRB no: B-1607/353-005). We collected the audio data of 39 infants who were assessed using seven multiple instruments, consisting of (1) ADOS, second edition (ADOS-2), (2) the autism diagnostic interview, revised (ADI-R), (3) the behavior development screening for toddlers interview (BeDevel-I), (4) the behavior development screening for toddlers play (BeDevel-P), (5) the Korean version of the childhood autism rating scale (K-CARS) re-named from CARS-2, (6) the social communication questionnaire (SCQ), and (7) the social responsiveness scale (SRS) [19-22]. The final diagnosis was based on the best clinical estimate diagnosis according to the DSM-5 ASD criteria by a licensed child psychiatrist using all of the available participant information. The participants' ages ranged between 6 and 24 months, where the average age was 19.20 months with a standard deviation (SD) of 2.52 months. Note here that the age means the age at the time when each infant visited the hospital to undergo an initial diagnosis examination. There were four males and six females diagnosed with ASD, whose average age was 14.72 months with a SD of 2.45. The remaining participants consisted of TD children (19 males and 10 females). Table 1 displays the collected data distribution, while Table 2 shows detailed information of collected data from the infants.

Table 1. Distribution of age and gender (male/female).
Ages (Month) 6-12 months 12-18 months 18-24 months
Age (average $\pm$ SD) No. of Subjects Diagnosed as ASD 0 1 M/3 F 3 M/3 F 19.20 $\pm$ 2.52
No. of Subjects Diagnosed as TD 5 M/1 F 14 M/9 F 0 14.72 $\pm$ 2.45
No. of Infant Subjects 5 M/1 F 15 M/12 F 3 M/3 F 15.92 $\pm$ 3.17

2.1. Data Collection and Acoustic Feature Extraction

{'page_number': 4,

'text': 'Table 2. Detailed information on the age, gender, and initial and definite diagnosis dates of each infant in Table 1.'

Infant ID	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39
Age (Months) on Initial Diagnosis	18	18	10	13	22	16	17	14	18	14	17	12	9	18	18	24	19	19	18	12	16	20	15	17	16	12	17	17	14	16	12	15	16	13	15	13	21	14	14
Gender	Male	Male	Male	Male	Female	Male	Female	Female	Male	Male	Female	Female	Male	Male	Female	Male	Male	Female	Male	Male	Female	Male	Male	Female	Male	Male	Female	Male	Male	Female	Male	Male	Female	Male	Male	Female	Male	Male	
Initial Diagnosis Date (Year/Month/Day)	2018/07/28	2017/07/27	2018/08/10	2017/06/10	2018/01/31	2018/03/17	2018/06/30	2018/01/06	2018/07/17	2017/11/04	2017/06/29	2018/01/20	2017/02/18	2017/03/04	2018/05/19	2018/08/08	2018/02/24	2017/04/18	2017/03/04	2016/12/31	2018/03/16	2017/10/14	2018/05/09	2017/02/04	2018/03/17														

2018/03/29 2017/01/25 2018/02/08 2018/01/13 2016/11/30 2017/03/22 2017/03/11
2017/12/05 2017/12/13 2017/03/25 2018/08/25 2017/06/24 2017/02/22 2018/01/27
Denite Final Diagnosis Date (Year/Month/Day) 2018/08/28 2017/08/27 2018/09/10
2017/07/10 2018/02/28 2018/04/17 2018/07/30 2018/02/06 2018/08/17 2017/12/04
2017/07/29 2018/02/20 2017/03/18 2017/04/04 2018/06/19 2018/09/08 2018/03/24
2017/05/18 2017/04/04 2017/01/31 2018/04/16 2017/11/14 2018/06/09 2017/03/04
2018/04/17 2018/04/29 2017/02/25 2018/03/08 2018/02/13 2016/12/30 2017/04/22
2017/04/11 2018/01/05 2018/01/13 2018/04/25 2018/09/25 2017/07/24 2017/03/22
2018/02/27 ASD/TD TD TD TD TD ASD TD TD TD TD TD ASD TD TD ASD TD ASD ASD ASD TD
TD TD ASD ASD TD TD TD TD ASD TD TD TD TD TD TD TD TD ASD TD TDAs each infant's
audio data were recorded during the clinical procedure to elicit behaviors from
infants, with the attendance of one doctor or clinician and one or both parents
with the child in the clinical area, the audio components consisted of various
speeches from the child, the clinician, and the parent(s), as well as noises
from toys or dragging chairs. Note here that the recordings were done in one of
two typical clinical rooms in SNUBH, where the room dimensions were 365 cm × 400
cm × 270 cm and 350 cm × 350 cm × 270 cm, and the hospital noise level was
around 40 dB. In order to analyze the vocal characteristics of the infants, each
audio clip was processed and split into audio segments containing the infant's
voice, not disturbed by music or clattering noises from toys or overlapped by
the voices of the clinician or parent(s). Each segment was classied into one of
ve categories, labeled from 0 to 4, for measuring the data distribution. Each
label was intended to show di erentiable characteristics relative to the
children's linguistic development: (1) 0 for one syllable, which is a short,'},
{'page_number': 5,
'text': "momentary single vocalization such as \"ah\" or \"ba\"; (2) 1 for two
syllables, commonly denoted as canonical babbling, as a reduplication of clear
babbling of two identical or variant syllables such as \"baba\" or \"baga\"; (3) 2
for babbling, not containing syllables; (4) 3 for rst word, such as \"mother\" or
\"father\"; and (5) 4 for atypical voice, including screaming or crying. The
distribution of each type of vocalization in seconds is shown in Table 3. The
number of vocalizations per category is presented along with a rational value
considering the di erence between the ASD and TD groups. While the data were
unbalanced and very small, the distribution of ASD and TD vocalizations show the
same tendency as reported in [10], where the ASD group showed a signi cantly
lower ratio of rst words and an increased ratio of atypical vocalizations,
revealing developmental delay in linguistic ability. Table 3. Amount (ratio) of
each type of vocalization in seconds. For acquiring quali ed and e ective feature
sets for the vocal data, eGeMAPS was employed for voice feature extraction.
GeMAPS is a popular feature set providing minimalistic speech features generally
utilized for automatic voice analysis rather than as a large brute force
parameter set. As an extended version, eGeMAPS contains 88 acoustic features
that were fully utilized in this experiment. Each recorded set of audio data
stored as a 48 kHz stereo le was down-sampled and down-mixed into a 16 kHz
mono-audio le, taking into consideration its usability and resolution in mel-
frequency cepstral coe cients (MFCCs). To extract the speech features for ASD
classi cation, each infant's utterances were segmented into 25 ms frames with a

10 ms overlap between frames. Then, 88 different features of the eGeMAPS were extracted for each frame with open source speech and music interpretation using the large-space extraction (OpenSMILE) toolkit [23], and these features were normalized by mean and standard deviation. The normalization scaling was acquired and fixed by normalizing the factors of the training data set. The features were grouped for each frame considering the time-relevant characteristics of the speech data.

2.2. Pre-Trained AE for Acoustic Features

To further process and refine the acoustic data, a feature-extracting AE was introduced. An AE is a hierarchical structure that is trained as a regression model for reproducing the input parameters. The AE takes inputs and converts them into latent representations, and then reconstructs the input parameters from the latent values [24]. If we consider an input of AE, $x \in R$, then the latent representation $z \in R^Y$ and the reconstruction of the input $y \in R^?$ are obtained by applying a nonlinear activation function f to the weight sum of z using a weighting matrix $W \in R^{4'}$ and a bias vector $b \in R^{d'}$, such as $WTx + b$ $z = f(1)WTz + b$ $y = fy = f(WTz + b')$ 2 where T is a matrix transpose operator. When the latent dimension $d' < d$, the output from the latent layer is considered to be a compressed, meaningful value extracted from the input, which is also noted as a bottleneck feature [25]. The normalized eGeMAPS features were applied to train the feature-extracting AE, applying the same data as the input and the target. The AE model contained a latent layer with a lowered, compacted feature dimension compared to the input layer to achieve the useful bottleneck feature.

Vocal Label	ASD	TD	0	1	2	3	4	Total
80.134	(0.104)	314.405	(0.409)	33.241	(0.043)	8.311	(0.011)	333.400
(0.433)	769.491	267.897	(0.250)	443.498	(0.414)	34.766	(0.032)	57.286
(0.054)	266.794	(0.249)	1070.241					

```
{'page_number': 6,
'text': '6 of 11The model was symmetrically structured, centering around the
latent layer, and the model could be divided into two components: the encoder,
consisting of layers from the input to the latent layers, and a decoder,
consisting of layers from the bottleneck to the output layers.The AE structure
is depicted in Figure 1. Our AE model consisted of FC layers, with the
dimensions of 88, 70, 54, 70, and 88 nodes for the input, hidden, latent,
hidden, and output layers, respectively. The hidden dimension was selected
experimentally and the bottleneck feature dimension was used for comparison with
previous research [17], where 54 features were selected considering the
statistical dissimilarity of the distributions between the ASD and TD features
based on the Mann-Whitney U test [26]. We additionally introduced an auxiliary
output as the binary categorical target for ASD and TD, which is known as the
semi-supervised method, to train the AE model effectively [27]. The auxiliary
output is depicted as Aux in Figure 1. The reconstructed features and auxiliary
classification can be written as  $z_i = f(W_{i-1,i}z_{i-1} + b_{i-1,i})$  (3)  $y_{rec} = W_{3,4}z_3 + b_{3,4}$   $y_{aux} = (W_{2,a}z_2 + b_{2,a})$  (4) (5) where  $y_{rec}$  refers to the reconstructed
eGeMAPS features,  $y_{aux}$  is the auxiliary classification result,  $f$  is the
activation function, and  $\sigma$  is the softmax activation.  $y_{rec} = W_{3,4}z_3 + b_{3,4}$   $y_{aux} = (W_{2,a}z_2 + b_{2,a})$ where theFigure 1. Structure of a semi-supervised auto-encoder
(AE) model. eGeMAPS, extended version of the Geneva minimalistic acoustic
parameter set; ASD, autism spectrum disorder; TD, typical development.The losses
```

of the reconstruction error for main AE target are measured using the mean absolute error, while the auxiliary ASD/TD target loss is the binary cross-entropy, and they are added and simultaneously optimized with rational hyper-parameters. The overall loss equation is $L_{total} = L_{recon} + L_{aux}$ where L_{recon} , L_{aux} , and L_{total} denote the reconstruction error, auxiliary loss using a binary cross-entropy loss function, and total loss, respectively. where $z_1 = f(W_0x + b_0)$, and $L_{aux} = -y_1 \log(y_1) - (1 - y_1) \log(1 - y_1)$ $L_{total} = L_{recon} + L_{aux}$,

{'page_number': 7,

'text': '7 of 11 For the proportion normalization we fetched training 2.3.

Establishing As the machine Each model classification supervised a positive We composed with 88 BLSTM prepared the SVMs five input activation, adaptive was controlled saving the of speech audio formerly, vocalization model was an SVM model, and Figure 2. Structure of a joint optimization model of an auto-encoder (AE) and bidirectional long short-term memory (BLSTM). The performance of each method was evaluated through 5-fold cross validation, where 95 average ASD utterances and 130 average TD utterances were proportionally distributed over 5. Performance Evaluation',

{'page_number': 8,

'text': '8 of 11 cases of vocalizations for the generalized estimation of unconcentrated utterance data. The averaged performances of the 5 validation splits of each model are described in Table 4. The labeled names of the BLSTM were used as the features for training the BLSTM model, where eGeMAPS-88 denotes 88 features of eGeMAPS, eGeMAPS-54 denotes 54 features selected by the Mann-Whitney U test, and AE-encoded denotes the joint optimized model. In the classification stage, one utterance was processed in the frame-wise method and the softmax output was converted to class indices 0 and 1, and if the average of class indices of the frames was over 0.5, then the utterance was considered an ASD child's utterance. The performances were scored with conventional measures, as well as unweighted average recall (UAR) and weighted average recall (WAR), chosen in the INTERSPEECH 2009 Emotion challenge, which considered imbalanced classes [34]. In the experiment, the SVM model showed very low precision, which was extremely biased toward the TD class. The BLSTM classifier with 88 features of eGeMAPS and the AE model showed considerable quality in terms of classifying ASD and TD children, while the AE model showed only marginal improvement in correctly classifying children with ASD compared to eGeMAPS-88. The 54 selected features showed degraded quality compared to eGeMAPS-88, obtaining more biased results toward children with TD. Table 4. Classification results from the support vector machine (SVM), BLSTM with 88 or 54 eGeMAPS features, 54 selected eGeMAPS features, and BLSTM with AE-encoded features. Models SVM BLSTM (eGeMAPS-54) BLSTM (eGeMAPS-88) BLSTM (AE-Encoded) Predicted To ASD TD ASD TD ASD TD ASD TD ASD TD
62 413 18 632 170 305 103 547 196 279 99 551 215 260 98 552 Accuracy Precision
Recall F1 score UAR 0.6178 0.1305 0.7750 0.2234 0.5514 0.6373 0.3579 0.6227
0.4545 0.5997 0.6640 0.4126 0.6644 0.5091 0.6302 0.6818 0.4526 0.6869 0.5457
0.6509 UAR, unweighted average recall. 4. Discussion The vanilla BLSTM model presented in [17] conducted discrimination on well-classified subjects with 10-month-old children and sorted 54 features from eGeMAPS that had a distinctive

distribution between ASD and TD selected by the Mann-Whitney U test using the three-fold cross-validation method. However, because the difference in the data distribution failed to achieve the same eGeMAPS feature selection between the test and classification results with the specified feature set presented herein, the application of an identical model structure and the adoption of the same feature domain will allow both approaches to be indirectly comparable. These results can be interpreted by the data distributions, and we performed t-stochastic neighbor embedding (t-SNE) analysis [35] on the training data set, which can nonlinearly squeeze the data dimension based on a machine learning algorithm. Figure 3 shows each data distribution as a two-dimensional scatter plot. In the figure, the eGeMAPS features from eGeMAPS-88 and eGeMAPS-54 showed almost identical distribution, except for the amount of ASD outliers, which implies that the ASD and TD features in the eGeMAPS features show similar distributions in this experiment. As shown in [16], eGeMAPS includes temporal features that are relevant to vocalizations and utterances; thus, these features might cause confusion regarding the discrimination between ASD and TD. The AE-encoded features, however, showed a redistributed feature map with a more characteristic distribution compared to the eGeMAPS features. This is because the AE-encoded features were compressed into a bottleneck feature, which was derived by weighting the matrix, paying attention to the significant parameters'},

{'page_number': 9,

'text': 'Sensors 2020, 20, 6762 9 of 11 while reducing the influence from the ambiguous parameters. While the joint optimization model achieved only marginally improved results compared to eGeMAPS-88, the distribution of the feature map would be more noticeable in improved feature extraction models, as well as more differentiable in complex models, although BLSTM with eight cells was employed for a comparison with conventional research in this experiment. Figure 3. Two-dimensional scatter plot for (a) eGeMAPS-88, (b) eGeMAPS-54, and (c) the AE processed by t-stochastic neighbor embedding (t-SNE). While the overall performance scores were comparably low for general classification problems on account of the subjectivity and complexity of problems, and the limitation in terms of the shortage of data, the results of the jointly optimized model imply the possibility of deep-learning-based feature extraction for the improvement of automated ASD/TD diagnosis under restricted circumstances. 5. Conclusions In this paper, we conducted experiments for discovering the possibility of auto-encoder-based feature extraction and a joint optimization method for the automated detection of atypicality in voices of children with ASD during early developmental stages. Under the condition of an insufficient and dispersed data set, the classification results were relatively poor in comparison to the general classification tasks based on deep learning. Although our investigation used a limited number of subjects and an unbalanced data set, the suggested auto-encoder-based feature extraction and joint optimization method revealed the possibility of feature dimension and a slight improvement in model-based diagnosis under such uncertain circumstances. In future work, we will focus on increasing the reliability of the proposed method by addition of a number of infants' speech data, refinement of the acoustic

features, an auto-encoder for feature extraction, and better, deeper, and up-to-date model structures. This research can also be extended to children with the age of 3 or 4 who can speak several sentences. In this case, we will investigate the linguistic features, as well as acoustic features, such as we have done in this paper. In addition to ASD detection, this research can be applied to the detection of infants with development delays. Author Contributions: All authors discussed the contents of the manuscript. H.K.K. contributed to the research idea and the framework of this study; G.B. and H.J.Y. provided the database and helped with the discussion; J.H.L. performed the experiments; G.W.L. contributed to the data collection and pre-processing. All authors have read and agreed to the published version of the manuscript. Funding: This work was supported by the Institute of Information & communications Technology Planning & evaluation (IITP) grant funded by the Korea government (MSIT) (No. 2019-0-00330, Development of AI Technology for Early Screening of Infant/Child Autism Spectrum Disorders based on Cognition of the Psychological Behavior and Response). Conflicts of Interest: The authors declare no conflict of interest.'}, {'page_number': 10, 'text': '10 of 11'}]}

```
[14]: # The dictionary is large and takes a long time to compute so we will save it_
      ↪ locally
import pickle

# Specify the folder path
folder_path = '/kaggle/working/'

# Save the list of dictionaries to a file in the specified folder
file_path = os.path.join(folder_path, 'pdf_and_text.pkl')
with open(file_path, 'wb') as f:
    pickle.dump(pdf_and_text, f)
```

Loading the saved list of dictionaries

```
[15]: with open('/kaggle/working/pdf_and_text.pkl', 'rb') as f:
      loaded_pdf_and_text = pickle.load(f)
```

0.0.1 Further text processing (splitting pages into sentences)

```
[16]: from spacy.lang.en import English

nlp = English()

# Add a sentencizer pipeline, see https://spacy.io/api/sentencizer/
nlp.add_pipe("sentencizer")

# Create a document instance as an example
doc = nlp("Hello, my name is Amit. This is my notebook related to RAG.")
assert len(list(doc.sents)) == 2
```

```
# Access the sentences of the document
list(doc.sents)
```

```
[16]: [Hello, my name is Amit., This is my notebook related to RAG.]
```

```
[17]: for pdf in loaded_pdf_and_text:
      for page in pdf['text']:
          page["sentences"] = list(nlp(page["text"]).sents)
          # Make sure all sentences are strings
          page["sentences"] = [str(sentence) for sentence in page["sentences"]]
          # Count the sentences
          page["page_sentence_count_spacy"] = len(page["sentences"])
```

```
[19]: loaded_pdf_and_text[5]
```

```
[19]: {'filename': '22_Ouss_ASD.pdf',
      'text': [{'page_number': 1,
                  'text': 'A R T I C L E O p e n A c c e s s Behavior and interaction imaging at
9 months of age predict autism/intellectual disability in high-risk infants with
West syndrome Lisa Ouss1, Giuseppe Palestra 2, Catherine Saint-Georges2,3,
Marluce Leitgel Gille1, Mohamed Afshar4, Hugues Pellerin2, Kevin Bailly2,
Mohamed Chetouani2, Laurence Robel1, Bernard Golse1, Rima Nabbout5, Isabelle
Desguerre5, Mariana Guergova-Kuras4 and David Cohen 2,3; , : ) ( 0 9 8 7 6 5 4 3
2 1; , : ) ( 0 9 8 7 6 5 4 3 2 1; , : ) ( 0 9 8 7 6 5 4 3 2 1; , : ) ( 0 9 8 7 6
5 4 3 2 1Abstract Automated behavior analysis are promising tools to overcome
current assessment limitations in psychiatry. At 9 months of age, we recorded 32
infants with West syndrome (WS) and 19 typically developing (TD) controls during
a standardized mother-infant interaction. We computed infant hand movements
(HM), speech turn taking of both partners (vocalization, pause, silences,
overlap) and motherese. Then, we assessed whether multimodal social signals and
interactional synchrony at 9 months could predict outcomes (autism spectrum
disorder (ASD) and intellectual disability (ID)) of infants with WS at 4 years.
At follow-up, 10 infants developed ASD/ID (WS+). The best machine learning
reached 76.47% accuracy classifying WS vs. TD and 81.25% accuracy classifying
WS+ vs. WS-. The 10 best features to distinguish WS+ and WS- included a
combination of infant vocalizations and HM features combined with synchrony
vocalization features. These data indicate that behavioral and interaction
imaging was able to predict ASD/ ID in high-risk children with
WS.IntroductionBehavior and interaction imaging is a promising domain of
affective computing to explore psychiatric conditions1-3. researchers have
Regarding attempted to identify reliable indicators of neurodevelop- mental
disorders (NDD) in high-risk populations (e.g., sib- lings of children with
autism) during the rst year of life to recommend early interventions4,5.
However, social signals and any alterations of them are very dif cult to
identify at such a young age6. In addition, exploring the quality and dynamics
of early interactions is a complex endeavor. It usually requires (i) the
```

perception and integration of mul- timodal social signals and (ii) an understanding of how two

Correspondence: Lisa Ouss (lisa.ouss@aphp.fr) or David Cohen (david.cohen@aphp.fr)

1Service de Psychiatrie de l'Enfant, AP-HP, Hôpital Necker, 149 rue de Sèvres, 75015 Paris, France

2Institut des Systèmes Intelligents et de Robotique, CNRS, UMR 7222, Sorbonne Université, 4 Place Jussieu, 75252 Paris Cedex, France

Full list of author information is available at the end of the article

interactive partners synchronize and proceed in turn taking^{7,8}. Affective computing offers the possibility to simulta- neously analyze the interaction of several partners while considering the multimodal nature and dynamics of social signals and behaviors⁹. To date, few seminal studies have attempted to social mother-infant interactions with or without a speci c condition, and these studies have focused on speech turns (e.g., Jaffe et al.¹⁰), motherese¹¹, head movements¹², hand movements¹³, movement facial expressions³. Here, we focused on West syndrome (WS), a rare epi- leptic encephalopathy with early onset (before age 1 year) and a high risk of NDD outcomes, including one-third of WS children showing later autism spectrum disorder (ASD) and/or intellectual disability (ID). We recruited 32 infants with WS and 19 typically developing (TD) controls to participate in a standardized early mother-infant

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'sentences': ['A R T I C L E O p e n A c c e s s Behavior and interaction imaging at 9 months of age predict autism/intellectual disability in high-risk infants with West syndrome Lisa Ouss¹, Giuseppe Palestra ², Catherine Saint-Georges^{2,3}, Marluce Leitgel Gille¹, Mohamed Afshar⁴, Hugues Pellerin², Kevin Bailly², Mohamed Chetouani², Laurence Robel¹, Bernard Golse¹, Rima Nabbout⁵, Isabelle Desguerre⁵, Mariana Guergova-Kuras⁴ and David Cohen ^{2,3}; , :) (0 9 8 7 6 5 4 3 2 1; , :) (0 9 8 7 6 5 4 3 2 1; , :) (0 9 8 7 6 5 4 3 2 1; , :) (0 9 8 7 6 5 4 3 2 1

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'We computed infant hand movements (HM), speech turn taking of both partners (vocalization, pause, silences, overlap) and motherese.',

'Then, we assessed whether multimodal social signals and interactional synchrony at 9 months could predict outcomes (autism spectrum disorder (ASD) and intellectual disability (ID)) of infants with WS at 4 years.',

'At follow-up, 10 infants developed ASD/ID (WS+).',

'The best machine learning reached 76.47% accuracy classifying WS vs. TD and 81.25% accuracy classifying WS+ vs. WS-. The 10 best features to distinguish WS+ and WS- included a combination of infant vocalizations and HM features combined with synchrony vocalization features.',

'These data indicate that behavioral and interaction imaging was able to predict ASD/ ID in high-risk children with WS. Introduction Behavior and interaction imaging is a promising domain of affective computing to explore psychiatric conditions¹⁻³.',

'researchers have Regarding attempted to identify reliable indicators of neurodevelop- mental disorders (NDD) in high-risk populations (e.g., sib- lings

of children with autism) during the first year of life to recommend early interventions^{4,5}.'

'However, social signals and any alterations of them are very difficult to identify at such a young age⁶.'

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'It usually requires (i) the perception and integration of multimodal social signals and (ii) an understanding of how two

Correspondence: Lisa Ouss (lisa.ouss@aphp.fr) or David Cohen (david.cohen@aphp.fr) 1Service de Psychiatrie de l'Enfant, AP-HP, Hôpital Necker, 149 rue de Sèvres, 75015 Paris, France 2Institut des Systèmes Intelligents et de Robotique, CNRS, UMR 7222, Sorbonne Université, 4 Place Jussieu, 75252 Paris Cedex, France Full list of author information is available at the end of the article

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'To date, few seminal studies have attempted to social mother-infant interactions with or without a specific condition, and these studies have focused on speech turns (e.g., Jaffe et al.¹⁰), motherese¹¹, head movements¹², hand movements¹³, movement facial expressions³. Here, we focused on West syndrome (WS), a rare epileptic encephalopathy with early onset (before age 1 year) and a high risk of NDD outcomes, including one-third of WS children showing later autism spectrum disorder (ASD) and/or intellectual disability (ID).'

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© The Author(s) 2020'],

'page_sentence_count_spacy': 13},

{'page_number': 2,

'text': 'interaction protocol and followed infants with WS to assess outcomes at 4 years of age. We aim to explore whether multimodal social signals and interpersonal infant-mother interactions at 9 months synchrony of could predict outcomes. Materials and methods Design, participants, and clinical measures We performed a prospective follow-up study of infants with WS¹⁴. The Institutional Review Board (Comité de Protection des Personnes from the Groupe-Hospitalier Necker Enfants Malades) approved the study, and both parents gave written informed consent after they received verbal and written information on the study. They were asked to participate to a follow-up study to assess outcome of WS taking into account development, early interaction, genetics and response to pharmacological treatment¹⁴. The study was conducted from November 2004 to March 2010 in the Neuro-pediatrics Department Center for Rare Epilepsia of Necker Enfants-Malades Hospital, Paris. Of the 41 patients screened during the study period, we enrolled all but two cases (N = 39) with WS. Seven patients dropped out before the age of 3 leading to a sample of 32 patients with detailed follow-up data. Typically developing infants (N = 19) were recruited from Maternal and Infant Prevention institutions, in pediatric consultations, or by proxy. To assess neurodevelopmental outcomes, we focused on ID and ASD. ID was assessed through the Brunet-Lézine Developmental Examination, performed for all children

at the age of 3 years. The Brunet-Lézine Developmental Examination estimates a developmental quotient (DQ) based upon normative data available for 3-year-old French toddlers¹⁵. The diagnosis of autism was based upon several measurements and an expert assessment that was blind to other variables: (i) At 3 years of age, all parents completed the Autism Diagnostic Interview- Revised (ADI-R) to assess autism signs by dimensions and developmental delay¹⁶. (ii) At 2 and 3 years of age, all patients were assessed with the Children's Autism Rating Scale (CARS)¹⁷. (iii) An expert clinician (LR) who was blind to child history assessed autism and ID from 20-min videotapes of child/mother play at 2 years of age. Finally, diagnoses of ASD and/or ID at age 4 were based upon a consensus approach using direct assessment of the child by a clinician with expertise in autism (LO) as well as by clinical information from the CARS, ADI-R, and DQ. Video recordings of infant-mother interactions were assessed between 9 and 12 months of age during a play session (Fig. 1). Two synchronized cameras (face and profile; Fig. S1A) recorded the movements in two dimensions while the infant was sitting in a baby chair. Audio interactions were also recorded. Around 9 months of age, we performed a laboratory recording of infant-mother interactions. We extracted audio features from the recordings and classified them as typical or atypical vocalization, pause, or non-vocalization. We then compared the results with the mother's vocalization, pause, or non-vocalization. We used a machine learning approach to classify the interactions as typical or atypical. We used a pipeline of our machine learning approach to classify WS vs. TD. Infant Mother Fig. 1 Pipeline of our machine learning approach to classify WS vs. TD.

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'1).',

'Two synchronized cameras (face and profile; Fig.',

'S1A) recorded the movements in two dimensions while the infant was sitting in a baby chair.',

'Audio interactions were also recorded around 9 months - At 4 years WS with Lab recording of infant-mother 10) - = ID/ASD interaction Audio features extraction vs. WS Infant: typical vocalization, without infant atypical vocalization, pause WS with Mother ID/ASD (N ID/ASD 32) Mother: vocalization, = motherese, pause (N Synchrony: overlap, silence, infant synchrony ratio 22) = = West syndrome children Machine 3 sequences of interaction learning WS without classification ID/ASD (N Free interaction 19) = = 19) Interaction = = - with the giraffe West syndrome controls (N: Typical developing Mother singing controls (N: vs. TD Typical developing Fig.',

'1 Pipeline of our machine learning approach to classify WS vs. TD. Infant Mother Fig.',

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'text': 'recorded. The standardized situation encompassed three sequences of 3 min: (sequence 1) free play after instructing the mother to interact "as usual" without any toy; (sequence 2) free play using the help of a toy (Sophie the giraffe); (sequence 3) mother singing to her baby. Due to the position of the baby chair on the floor and the mother's seated position, the mother was positioned slightly higher in all of the recordings. The mother's indicated position was on the left of the child as shown on the picture, but exceptions were sometimes observed during the recordings. For infant hand movement (HM) features, 1 min was extracted from each 3-min video and all recordings,

according to two criteria: the child's hands should be visible for at least part of the sequence (e.g., the mother is not leaning on the child), and the minute represented the greatest amount of interaction between the mother and the child. For audio and speech turn-taking computing, we only used the 3-min audio recording of sequence 1. Vision computing (Fig. S1B, vision computing panel) To process infant hand movements (HM), we used the methods developed in Ouss et al. 13. Here, we summarize the successive steps to calculate HM features. In step 1 (hand trajectory extraction and data processing), the two-dimensional coordinates of the hand were extracted from each of the video recordings by tracking a wristband on the right hand (yellow in Fig. S1A, video-audio recording panel). The tracking framework comprised three steps: prediction, observation, and estimation as proposed in ref. 18. As the hand motion was highly nonlinear, we developed an approach using a bootstrap-based particle filter with a first-order model to address abrupt changes in direction and speed 19,20. To address hand occlusion, we implemented an approach combining tracking with detection by adding a boolean variable to the state vector associated with each particle 18. Each extracted trajectory consisted of 1500 pairs of x and y coordinates (25 frames per second, generating 1500 pairs of coordinates in the 60 s; see Fig. S1 left panel, vision computing). The frames where the hand was not visible were clearly indicated in each trajectory as missing coordinates for these time points. To account for differences in the camera zoom parameters, the trajectories obtained were normalized using a fixed reference system present in the settings of each video recording. The normalization was performed on all trajectories, and 95% of the normalization factors ranged between 0.8 and 1.22 with a few outlier trajectories that required greater correction. Forty-one percent of the trajectories required <5% correction. Although the recordings between the two cameras were synchronized and in principle allowed 3D reconstruction of the trajectory, the accumulation of missing data prevented such reconstruction. However, 2D motion capture with appropriately defined movement descriptors can be powerful for detecting clinically relevant changes 21, thereby justifying the independent analysis of the 2D-trajectory videos (see Fig. S1B, vision computing, 2d panel on the left). In step 2, the descriptors of the HM were calculated from the planar trajectories (Fig. S1B, table shown in the vision computing panel). Descriptors covered those already reported in the literature as important in characterizing infants' HM 21. (1) To describe the space explored by the hand, we calculated the maximum distance observed on the two axes (xRange, yRange) and the standard deviation of the X and Y coordinates observed during the 60 s (xSd, ySd). We also calculated the maximum distance between any two points of the trajectory using (<http://algs4.cs.princeton.edu/code/>) (Fig. S1B, vision computing panel, red line in the third panel from the left). (2) To evaluate HM dynamics, we calculated the velocity and acceleration. (3) Also related to HM dynamics, we calculated HM pauses defined as part of the trajectory in which the velocity was lower than a specific threshold for a minimum duration of 4 s. (4) Finally, the curvature of the trajectories was calculated using a standard definition of the curvature (κ) of plane curves in Cartesian coordinates as $\kappa(t) = \sqrt{x''(t)^2 + y''(t)^2}$. The curvature calculated at each point of

the trajectory is presented in the right panel of Fig. S1B (video computing), where the first 1.2 s of the trajectory are plotted and the associated calculated curvatures at each point (and respective time, indicated on the axis) are presented as columns. Audio computing (Fig. S1C, audio computing) We extracted two types of audio social signals from the audio channel of the mother-infant interaction: speech turn taking (STT) and motherese. For STT extraction, we followed the methods developed by Weisman et al.²² and Bourvis et al.²³ (Fig. S1, audio computing). First, we used ELAN to segment the infants' and mothers' speech turns and annotate the dialog acts. Mothers' audio interactions were categorized as mother vocalization (meaningful laugh, singing, animal sounds) or other vocalizations, noise (clap hands, snap fingers or snap the tongue, mouth infants' audio production was noise, etc.). Similarly, defined as infant vocalization (babbling vocalizations, laugh, and cry) or atypical vocalization (other noise such as "rattle"). The infant's and mother's utterances were labeled by two annotators (blind to group status). Cohen's kappa between the two annotators was calculated for each dyad, each task and each item of the grid. For all items, the kappa values were between 0.82 and 1. From the annotation, we extracted all the speech turns of the infant and the mother. A speech turn is a continuous stream of speech with <150 ms of silence. We',

'sentences': ['recorded.'],

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 'text': 'obtained a list of triples: speaker label (infant or mother), start time, and duration of speech turn. From these triples, we also deduced the start time and duration of the time segments when the mother or the infant were not speaking (pauses). Therefore, we extracted Mother Vocalizations; Mother Other Noise; Infant Vocalizations; Infant Infant Atypical Vocalizations; Mother Pauses; Pauses. We also extracted three dyadic features: (1) Silence dened as sequences of time during which neither participant was speaking for more than 150 ms; (2) Overlap Ratio dened as the duration of vocalization overlaps between mothers and infants divided by the duration of the total interaction. This ratio measures the proportion of interactional time in which both participants were simultaneously vocalizing; (3) Infant Syn- chrony Ratio dened as the number of infants' responses to their mother's vocalization within a time limit of 3 s divided by the number of mother vocalizations during the time paradigm. The 3-s window was based on the avail- able literature on synchrony^{7,24}.From the mother vocalizations, we also computed affective speech analysis, as previous work has shown that motherese may shape parent-infant interactions²⁵. The segments of mother vocalizations were analyzed using a computerized classier for categorization as "motherese" or "non-motherese/other speech" initially developed to analyze home movies¹¹. The system exploits the fusion of two classiers, namely, segmental and suprasegmental²⁶. Consequently, the utterances are characterized by both frequency cepstrum coef cients) and segmental suprasegmental/prosodics (e.g., statistics with regard to fundamental frequency,

energy, and duration) features. The detector used the GMM (Gaussian mixture model) classifier for both segmental and suprasegmental features (M, number of Gaussians for the GMM Classifier: M = 12 and 15, respectively, and α = weighting coefficient used in the equation fusion: $\alpha = 0.4$). For the purpose of the current study, we explored the performance of our motherese classifier in French mothers. We analyzed 200 sequences from French mothers (100 motherese vs. 100 other speech) that were blindly validated by two psycholinguists. We calculated the Intraclass correlation (ICC) between the two raters (the expert and the algorithm) and found a good and very significant ICC (ICC = 0.79 (95% CI: 0.59-0.90), $p < 0.001$). This level of prediction made it suitable for further analysis of the entire data set. Based on this automatic detection of motherese, we created two subclasses for mother vocalizations: motherese vs. non-motherese. Two variables were derived: Motherese Ratio (duration of motherese vocalization/duration of interaction) and Non-motherese Ratio (duration of non-motherese vocalization/duration of interaction). We also derived two synchrony ratios: SynchronyMotherese Ratio and Synchrony Non-motherese Ratio, which reflect the ratio of time during which the infant vocalizes in response to his/her mother motherese and other speech (non-motherese). Prediction of the outcome using machine learning

The pipeline of our approach is shown in Fig. 1. First, a data quality analysis was performed to ensure the validity of the data. As expected, all data were available for audio analysis. However, a substantial proportion of the data were discarded due to video recording or vision computing issues. We finally kept 18 video recordings for the WS and 17 videos for the TD groups. Second, given the number of features (21 infant HM for each camera and each sequence; 16 STT) compared with the data set (32 WS and 19 TD), we reduced our data using principal component analysis (PCA). Third, we tested several algorithms to classify WS vs. TD based on the whole data set available for both vision and audio computing features (leave one out) (Table S1). The best algorithm was decision stump²⁷. All results presented here are based on the classification with a decision stump algorithm. We also analyzed WS with ID/ASD (WS+) vs. WS without ID/ASD (WS-). For each classification, we also extracted a confusion matrix and explored which individual features contributed the most to a given classification using Pearson correlations. Results Table S2 summarizes the demographic and clinical characteristics of children with WS. At follow-up, 10 infants out of 32 children with WS developed ASD/ID (WS+). Eight children had ASD and ID, whereas 2 had only ID. As expected, all variables related to ASD and ID were significantly different in WS+ compared with WS-. Figure 2a summarizes the best classification models using the decision stump algorithm (leave one out). As shown, multimodal classification outperformed unimodal classification to distinguish WS and TD. Therefore, we only used the multimodal approach to classify WS+ vs. WS-. The best model reached 76.47% accuracy classifying WS vs. TD and 81.25% accuracy classifying WS+ vs. WS- based on multimodal features extracted during early interactions. Interestingly, the confusion matrices (Fig. 2b) show that when classifying WS vs. TD, all errors came from TD being misclassified as WS (N = 12); when classifying WS+ vs. WS-, most errors came from WS+ being misclassified as WS- (N = 5). Table 1 lists the best features for each multimodal classification based on the Pearson correlation values. The best

features to distinguish WS and TD included four infant HM features, 1 mother audio feature. In contrast, the best features to distinguish WS+ and WS- included a (N = 2), combination of',

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'text': 'synchrony vocalization features (N = 3) and infant HM features (N = 5), the last of which showed lower correlation scores. Discussion To the best of our knowledge, this is the first study to apply multimodal to mother-infant interactions in the context of WS. Combining an infant-mother interaction at 9 months significantly predicted the development of ASD or severe to moderate ID at 4 years of age in the high-risk children with WS. Confusion matrices showed that the classification errors were not random, enhancing the interest of the computational method proposed here. In addition, the best contributing features for the performed classifications differed when classifying WS vs. TD and WS+ vs. WS-. Infant HMs were the most significant features to distinguish WS versus TD, probably reflecting the motor impact due to acute WS encephalopathy. For classifying WS+ vs. WS-, the contribution of infant audio features and synchrony features became much more relevant combined with several HM features. We believe that the importance of synchrony and reciprocity during early interactions is in line with recent studies that have investigated the risk of ASD or NDD during the first year of life from home movies (e.g., refs. 11,24), from prospective follow-up of high-risk infants such as siblings (e.g., refs. 4,28) or infants with WS (e.g., ref. 14), and from prospective studies assessing tools to screen risk for autism (e.g., ref. 29). In the field of ASD, synchrony, reciprocity, parental sensitivity, and emotional engagement are now proposed as targets of early interventions³⁰, which could prevent early interactive vicious circles. Parents of at-risk infants try to compensate for the lack of interactivity of their child by modifying their stimulation and therefore sometimes interactions²⁴. Early reinforcing identification of these interactive targets is especially useful among babies with neurological comorbidities because delays and impairments in early social interactions are not sufficient to predict ASD. Similarly, we believe that the importance of HM in distinguishing WS vs. TD on one hand, and WS+ vs. WS- on the other hand, is also in line with the studies that investigated the importance of non-social behaviors for investigating the risk of ASD or NDD during the first year of life.

For example, studying home movies, Purpura et al. found more bilateral HM and nger movements in infants who will later develop ASD³¹. Similarly, several prospective follow-up studies of high-risk siblings³²⁻³⁵ or retrospective studies on home movies^{36,37} reported specific motor atypical repertoire in infants with ASD. In ASD, early social signals have previously been assessed with automatized and computational procedures, focusing on eye tracking at early stages³⁸⁻⁴⁰, vocal productions⁴¹, analysis of acoustics of first utterances or cry episodes⁴², but none was done in an interactive setting. Our study proposed a paradigm shift from the assessment of infant behavior to dyadic assessment of interactions, as previously achieved in retrospective approaches using home movies²⁴. The aim is not to implement studies of social signal processing in routine clinical work but rather to decompose clinical intuitions and signs and validate the most relevant cues of these clinical features. From clinical work, back to clinics, social signal processing is a rigorous step to help clinicians better identify and assess early targets of interventions. Given the exploratory nature of both our approach and method, our results should be interpreted with caution taking into account strengths (prospective follow-up, automatized multimodal social signal processing, and ecological standardized assessment) and limitations. These limitations include (1) the overall sample size knowing that WS is a rare disease; (2) the high rate of missing data during video recording due to the ecological conditions of the infant-mother interaction (mothers interposing between the camera and the infant); the small sample size of WS+ (N = 10) that limited the power of machine learning methods. We conclude that the method proposed here combining multimodal automatized assessment of social signal processing during early interaction with infants at risk for NDD is a promising tool to decipher clinical features that remain difficult to identify and assess. In the context of WS, we showed that such a method we proposed to label 'behavioral and interaction imaging' was able to significantly predict the development of ASD or ID at 4 years of age in high-risk children who had WS and were assessed at 9 months of age.

Acknowledgements The authors thank all of the patients and families who participated in this study. The study was funded by the EADS foundation (PILE), by the Agence Nationale de la Recherche (ANR-12-SAMA-006-1) and the Groupement de Recherche en Psychiatrie (GDR-3557). It was partially performed in the Labex SMART (ANR-11-LABX-65), which is supported by French state funds and managed by the ANR in the Investissements d'Avenir program under reference ANR-11-IDEX-0004-02. The sponsors had no involvement in the study design, data analysis, or interpretation of the results.

Author details 1Service de Psychiatrie de l'Enfant, AP-HP, Hôpital Necker, 149 rue de Sèvres, 75015 Paris, France. 2Institut des Systèmes Intelligents et de Robotique, CNRS, UMR 7222, Sorbonne Université, 4 Place Jussieu, 75252 Paris Cedex, France. 3Département de Psychiatrie de l'Enfant et de l'Adolescent, AP-HP, Hôpital Pitié-Salpêtrière, 47-83, Boulevard de l'Hôpital, 75651 Paris, Cedex 13, France. 4Ariana Pharmaceuticals, Research Department, Paris, France. 5Service de Neuropédiatrie, AP-HP, Hôpital Necker, 136, Rue de Vaugirard, 75015 Paris, France',

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'In addition, the best contributing features for the performed classi- cations differed when classifying WS vs. TD and WS+ vs. WS-. Infant HMs were the most signi- cant features to distin- guish WS versus TD, probably re- ecting the motor impact due to acute WS encephalopathy.',

'For classifying WS+ vs. WS-, the contribution of infant audio features and synchrony features became much more relevant combined with several HM features.',

'We believe that the importance of synchrony and reciprocity during early interactions is in line with recent studies that have investigated the risk of ASD or NDD during the rst year of life from home movies (e.g., refs.',

'11,24), from prospective follow-up of high-risk infants such as siblings (e.g., refs.',

'4,28) or infants with WS (e.g., ref.',

'14), and from prospective studies assessing tools to screen risk for autism (e.g., ref.',

'29).',

'In the eld of ASD, synchrony, reciprocity, parental sensitivity, and emotional engagement are now proposed as targets of early interventions³⁰, which could prevent early inter- active vicious circles.',

'Parents of at-risk infants try to compensate for the lack of interactivity of their child by modifying their stimulation and therefore sometimes interactions²⁴.',

'Early reinforcing identi- cation of these interactive targets is especially useful among babies with neurological comorbidities because delays and impairments in early social interactions are not suf- cient to predict ASD. Similarly, we believe that the importance of HM in distinguishing WS vs. TD on one hand, and WS+ vs. WS- on the other hand, is also in line with the studies that investigated the importance of non-social behaviors for investigating the risk of ASD or NDD during the rst year of life.',

'For example, studying home movies, Purpura et al.',

'found more bilateral HM and nger movements in infants who will later develop ASD³¹.',

'Similarly, several prospective follow-up studies of high-risk siblings³²⁻³⁵ or retrospective studies on home movies^{36,37} reported spe- cific motor atypical repertoire in infants with ASD. In ASD, early social signals have previously been assessed with automatized and computational procedures, focusing on eye tracking at early stages³⁸⁻⁴⁰, vocal pro- ductions⁴¹, analysis of acoustics of rst utterances or cry episodes⁴², but none was done in an interactive setting.',

'Our study proposed a paradigm shift from the assessment of infant behavior to dyadic assessment of interactions, as previously achieved in retrospective

approaches using home movies²⁴.' ,

'The aim is not to implement studies of social signal processing in routine clinical work but rather to decompose clinical intuitions and signs and validate the most relevant cues of these clinical features.' ,

'From clinical work, back to clinics, social signal processing is a rigorous step to help clinicians better identify and assess early targets of interventions.' ,

'Given the exploratory nature of both our approach and method, our results should be interpreted with caution taking into account strengths (prospective follow-up, automatized multimodal social signal processing, and ecological standardized assessment) and limitations.' ,

'These limitations include (1) the overall sample size knowing that WS is a rare disease; (2) the high rate of missing data during video recording due to the ecological conditions of the infant-mother interaction (mothers interposing between the camera and the infant); the small sample size of WS+ (N = 10) that limited the power of machine learning methods.' ,

'We conclude that the method proposed here combining multimodal automatized assessment of social signal processing during early interaction with infants at risk for NDD is a promising tool to decipher clinical features that remain difficult to identify and assess.' ,

'In the context of WS, we showed that such a method we proposed to label 'behavioral and interaction imaging' was able to significantly predict the development of ASD or ID at 4 years of age in high-risk children who had WS and were assessed at 9 months of age.' ,

'Acknowledgements The authors thank all of the patients and families who participated in this study.' ,

'The study was funded by the EADS foundation (PILE), by the Agence Nationale de la Recherche (ANR-12-SAMA-006-1) and the Groupement de Recherche en Psychiatrie (GDR-3557).' ,

'It was partially performed in the Labex SMART (ANR-11-LABX-65), which is supported by French state funds and managed by the ANR in the Investissements d'Avenir program under reference ANR-11-IDEX-0004-02.' ,

'The sponsors had no involvement in the study design, data analysis, or interpretation of the results.' ,

'Author details ¹Service de Psychiatrie de l'Enfant, AP-HP, Hôpital Necker, 149 rue de Sèvres, 75015 Paris, France.' ,

'²Institut des Systèmes Intelligents et de Robotique, CNRS, UMR 7222, Sorbonne Université, 4 Place Jussieu, 75252 Paris Cedex, France.' ,

'³Département de Psychiatrie de l'Enfant et de l'Adolescent, AP-HP, Hôpital Pitié-Salpêtrière, 47-83, Boulevard de l'Hôpital, 75651 Paris, Cedex 13, France.' ,

'⁴Ariana Pharmaceuticals, Research Department, Paris, France.' ,

'⁵Service de Neuropédiatrie, AP-HP, Hôpital Necker, 136, Rue de Vaugirard, 75015 Paris, France'] ,

'page_sentence_count_spacy': 33} ,

{'page_number': 7 ,

'text': 'Conflict of interest The authors declare that they have no conflict of

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interest.Publisher's note Springer Nature remains neutral with regard to
jurisdictional claims in published maps and institutional
affiliations.Supplementary Information accompanies this paper at
(https://doi.org/10.1038/s41398-020-0743-8).Received: 7 December 2019 Revised:
13 January 2020 Accepted: 16 January 2020Published online: 03 February 2020',
    'sentences': ['Conict of interest The authors declare that they have no
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    'Publisher's note Springer Nature remains neutral with regard to
jurisdictional claims in published maps and institutional affiliations.',
    'Supplementary Information accompanies this paper at (https://doi.org/10.1038/s41398-020-0743-8).Received: 7 December 2019 Revised: 13 January 2020
Accepted: 16 January 2020Published online: 03 February 2020'],
    'page_sentence_count_spacy': 3]]}

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0.0.2 Chunking our sentences together

```

[20]: # Define split size to turn groups of sentences into chunks
num_sentence_chunk_size = 10

# Create a function that recursively splits a list into desired sizes
def split_list(input_list: list,
               slice_size: int)-> list[list[str]]:
    return [input_list[i:i + slice_size] for i in range(0,
    ↪len(input_list),slice_size)]

[21]: # Loop through pages and texts and split sentences into chunks
for pdf in loaded_pdf_and_text:
    for page in pdf['text']:
        page["sentence_chunks"] = split_list(input_list=page["sentences"],
                                              slice_size=num_sentence_chunk_size)
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9 months of age predict autism/intellectual disability in high-risk infants with
West syndrome Lisa Ouss1, Giuseppe Palestra 2, Catherine Saint-Georges2,3,
Marluce Leitgel Gille1, Mohamed Afshar4, Hugues Pellerin2, Kevin Bailly2,
Mohamed Chetouani2, Laurence Robel1, Bernard Golse1, Rima Nabbout5, Isabelle
Desguerre5, Mariana Guergova-Kuras4 and David Cohen 2,3; , : ) ( 0 9 8 7 6 5 4 3
2 1; , : ) ( 0 9 8 7 6 5 4 3 2 1; , : ) ( 0 9 8 7 6 5 4 3 2 1; , : ) ( 0 9 8 7 6
5 4 3 2 1Abstract Automated behavior analysis are promising tools to overcome
current assessment limitations in psychiatry. At 9 months of age, we recorded 32
infants with West syndrome (WS) and 19 typically developing (TD) controls during
a standardized mother-infant interaction. We computed infant hand movements

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(HM), speech turn taking of both partners (vocalization, pause, silences, overlap) and motherese. Then, we assessed whether multimodal social signals and interactional synchrony at 9 months could predict outcomes (autism spectrum disorder (ASD) and intellectual disability (ID)) of infants with WS at 4 years. At follow-up, 10 infants developed ASD/ID (WS+). The best machine learning reached 76.47% accuracy classifying WS vs. TD and 81.25% accuracy classifying WS+ vs. WS-. The 10 best features to distinguish WS+ and WS- included a combination of infant vocalizations and HM features combined with synchrony vocalization features. These data indicate that behavioral and interaction imaging was able to predict ASD/ ID in high-risk children with WS.

Introduction Behavior and interaction imaging is a promising domain of affective computing to explore psychiatric conditions¹⁻³. researchers have attempted to identify reliable indicators of neurodevelopmental disorders (NDD) in high-risk populations (e.g., siblings of children with autism) during the first year of life to recommend early interventions^{4,5}. However, social signals and any alterations of them are very difficult to identify at such a young age⁶. In addition, exploring the quality and dynamics of early interactions is a complex endeavor. It usually requires (i) the perception and integration of multimodal social signals and (ii) an understanding of how two

Correspondence: Lisa Ouss (lisa.ouss@aphp.fr) or David Cohen (david.cohen@aphp.fr) ¹Service de Psychiatrie de l'Enfant, AP-HP, Hôpital Necker, 149 rue de Sèvres, 75015 Paris, France ²Institut des Systèmes Intelligents et de Robotique, CNRS, UMR 7222, Sorbonne Université, 4 Place Jussieu, 75252 Paris Cedex, France Full list of author information is available at the end of the article

interactive partners synchronize and proceed in turn taking^{7,8}. Affective computing offers the possibility to simultaneously analyze the interaction of several partners while considering the multimodal nature and dynamics of social signals and behaviors⁹. To date, few seminal studies have attempted to social mother-infant interactions with or without a specific condition, and these studies have focused on speech turns (e.g., Jaffe et al.¹⁰), motherese¹¹, head movements¹², hand movements¹³, movement facial expressions³. Here, we focused on West syndrome (WS), a rare epileptic encephalopathy with early onset (before age 1 year) and a high risk of NDD outcomes, including one-third of WS children showing later autism spectrum disorder (ASD) and/or intellectual disability (ID). We recruited 32 infants with WS and 19 typically developing (TD) controls to participate in a standardized early mother-infant

© The Author(s) 2020',

'sentences': ['A R T I C L E O p e n A c c e s s Behavior and interaction imaging at 9 months of age predict autism/intellectual disability in high-risk infants with West syndrome Lisa Ouss¹, Giuseppe Palestra ², Catherine Saint-Georges^{2,3}, Marluce Leitgel Gille¹, Mohamed Afshar⁴, Hugues Pellerin², Kevin Bailly², Mohamed Chetouani², Laurence Robel¹, Bernard Golse¹, Rima Nabbout⁵, Isabelle Desguerre⁵, Mariana Guergova-Kuras⁴ and David Cohen ^{2,3}; , :) (0 9 8 7 6 5 4 3 2 1; , :) (0 9 8 7 6 5 4 3 2 1; , :) (0 9 8 7 6 5 4 3 2 1; , :) (0 9 8 7 6 5 4 3 2 1Abstract Automated behavior analysis are promising tools to overcome current assessment limitations in psychiatry.',

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'Then, we assessed whether multimodal social signals and interactional synchrony at 9 months could predict outcomes (autism spectrum disorder (ASD) and intellectual disability (ID)) of infants with WS at 4 years.',

'At follow-up, 10 infants developed ASD/ID (WS+).',

'The best machine learning reached 76.47% accuracy classifying WS vs. TD and 81.25% accuracy classifying WS+ vs. WS-. The 10 best features to distinguish WS+ and WS- included a combination of infant vocalizations and HM features combined with synchrony vocalization features.',

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recording panel). The tracking framework comprised three steps: prediction, observation, and estimation as proposed in ref. 18. As the hand motion was highly nonlinear, we developed an approach using a bootstrap-based particle filter with a first-order model to address abrupt changes in direction and speed^{19,20}. To address hand occlusion, we implemented an approach combining tracking with detection by adding a boolean variable to the state vector associated with each particle¹⁸. Each extracted trajectory consisted of 1500 pairs of x and y coordinates (25 frames per second, generating 1500 pairs of coordinates in the 60 s; see Fig. S1 left panel, vision computing). The frames where the hand was not visible were clearly indicated in each trajectory as missing coordinates for these time points. To account for differences in the camera zoom parameters, the trajectories obtained were normalized using a fixed reference system present in the settings of each video recording. The normalization was performed on all trajectories, and 95% of the normalization factors ranged between 0.8 and 1.22 with a few outlier trajectories that required greater correction. Forty-one percent of the trajectories required <5% correction. Although the recordings between the two cameras were synchronized and in principle allowed 3D reconstruction of the trajectory, the accumulation of missing data prevented such reconstruction. However, 2D motion capture with appropriately defined movement descriptors can be powerful for detecting clinically relevant changes²¹, thereby justifying the independent analysis of the 2D-trajectory videos (see Fig. S1B, vision computing, 2d panel on the left). In step 2, the descriptors of the HM were calculated from the planar trajectories (Fig. S1B, table shown in the vision computing panel). Descriptors covered those already reported in the literature as important in characterizing infants' HM²¹. (1) To describe the space explored by the hand, we calculated the maximum distance observed on the two axes (xRange, yRange) and the standard deviation of the X and Y coordinates observed during the 60 s (xSd, ySd). We also calculated the maximum distance between any two points of the trajectory using (<http://algs4.cs.princeton.edu/code/>) (Fig. S1B, vision computing panel, red line in the third panel from the left). (2) To evaluate HM dynamics, we calculated the velocity and acceleration. (3) Also related to HM dynamics, we calculated HM pauses defined as part of the trajectory in which the velocity was lower than a specific threshold for a minimum duration of 4 s. (4) Finally, the curvature of the trajectories was calculated using a standard definition of the curvature ($\kappa(t)$) of plane curves in Cartesian coordinates as $\kappa(t) = \sqrt{x'(t)^2 + y'(t)^2}$. The curvature calculated at each point of the trajectory is presented in the right panel of Fig. S1B (video computing), where the first 1.2 s of the trajectory are plotted and the associated calculated curvatures at each point (and respective time, indicated on the axis) are presented as columns. Audio computing (Fig. S1C, audio computing) We extracted two types of audio social signals from the audio channel of the mother-infant interaction: speech turn taking (STT) and motherese. For STT extraction, we followed the methods developed by Weisman et al.²² and Bourvis et al.²³ (Fig. S1, audio computing). First, we used ELAN to segment the infants' and mothers' speech turns and annotate the dialog acts. Mothers' audio interactions were categorized as mother vocalization (meaningful laugh, singing, animal sounds) or

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sequences of time during which neither participant was speaking for more than
150 ms; (2) Overlap Ratio dened as the duration of vocalization overlaps
between mothers and infants divided by the duration of the total interaction.
This ratio measures the proportion of interactional time in which both partici-
pants were simultaneously vocalizing; (3) Infant Syn- chrony Ratio dened as the
number of infants' responses to their mother's vocalization within a time limit
of 3 s divided by the number of mother vocalizations during the time paradigm.
The 3-s window was based on the avail- able literature on synchrony7,24. From the
mother vocalizations, we also computed affective speech analysis, as previous
work has shown that motherese may shape parent-infant interactions25. The
segments of mother vocalizations were analyzed using a computerized classier
for categorization as "motherese" or "non-motherese/other speech" initially
developed to analyze home movies11. The system exploits the fusion of two
classiers, namely, segmental and suprasegmental26. Consequently, the utterances
are characterized by both frequency cepstrum coef cients) and segmental
suprasegmental/prosodics (e.g., statistics with regard to fundamental frequency,
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study, we explored the performance of our motherese classifier in French mothers. We analyzed 200 sequences from French mothers (100 motherese vs. 100 other speech) that were blindly validated by two psycholinguists. We calculated the Intraclass correlation (ICC) between the two raters (the expert and the algorithm) and found a good and very significant ICC ($ICC = 0.79$ (95% CI: 0.59-0.90), $p < 0.001$). This level of prediction made it suitable for further analysis of the entire data set. Based on this automatic detection of motherese, we created two subclasses for mother vocalizations: motherese vs. non-motherese. Two variables were derived: Motherese Ratio (duration of motherese vocalization/duration of interaction) and Non-motherese Ratio (duration of non-motherese vocalization/duration of interaction). We also derived two synchrony ratios: SynchronyMotherese Ratio and Synchrony Non-motherese Ratio, which reflect the ratio of time during which the infant vocalizes in response to his/her mother motherese and other speech (non-motherese). Prediction of the outcome using machine learning

The pipeline of our approach is shown in Fig. 1. First, a data quality analysis was performed to ensure the validity of the data. As expected, all data were available for audio analysis. However, a substantial proportion of the data were discarded due to video recording or vision computing issues. We finally kept 18 video recordings for the WS and 17 videos for the TD groups. Second, given the number of features (21 infant HM for each camera and each sequence; 16 STT) compared with the data set (32 WS and 19 TD), we reduced our data using principal component analysis (PCA). Third, we tested several algorithms to classify WS vs. TD based on the whole data set available for both vision and audio computing features (leave one out) (Table S1). The best algorithm was decision stump²⁷. All results presented here are based on the classification with a decision stump algorithm. We also analyzed WS with ID/ASD (WS+) vs. WS without ID/ASD (WS-). For each classification, we also extracted a confusion matrix and explored which individual features contributed the most to a given classification using Pearson correlations. Results Table S2 summarizes the demographic and clinical characteristics of children with WS. At follow-up, 10 infants out of 32 children with WS developed ASD/ID (WS+). Eight children had ASD and ID, whereas 2 had only ID. As expected, all variables related to ASD and ID were significantly different in WS+ compared with WS-. Figure 2a summarizes the best classification models using the decision stump algorithm (leave one out). As shown, multimodal classification outperformed unimodal classification to distinguish WS and TD. Therefore, we only used the multimodal approach to classify WS+ vs. WS-. The best model reached 76.47% accuracy classifying WS vs. TD and 81.25% accuracy classifying WS+ vs. WS- based on multimodal features extracted during early interactions. Interestingly, the confusion matrices (Fig. 2b) show that when classifying WS vs. TD, all errors came from TD being misclassified as WS ($N = 12$); when classifying WS+ vs. WS-, most errors came from WS+ being misclassified as WS- ($N = 5$). Table 1 lists the best features for each multimodal classification based on the Pearson correlation values. The best features to distinguish WS and TD included four infant HM features, 1 mother audio feature. In contrast, the best features to distinguish WS+ and WS- included a ($N = 2$), combination of,

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'This ratio measures the proportion of interactional time in which both participants were simultaneously vocalizing; (3) Infant Synchrony Ratio defined as the number of infants' responses to their mother's vocalization within a time limit of 3 s divided by the number of mother vocalizations during the time paradigm.',

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one out) matrices I West vs. TD (n=51) 90 Clas: ied West T 80 as D West 32 70 el
12 60 50 West with ID/ASD vs. 40 I West with out ID/ASD 30 (N=32) West with West
with 20 ID/ASD out ID/ASD West with 5 5 10 ID/ASD West with 1 21 Mutimodal Video
Audio out ID/ASDFig. 2 Machine learning classiation of WS vs. TD and WS+ vs.
WS- based on uni- and multimodal features extracted during early infant-mother
interaction.Table 1 Best features for classiation (based on signi cant
Pearson's correlation between feature and class).Feature characteristics Pearson
r West vs. Typical developing Ratio of all maternal audio intervention during
free interaction Audio, mother 0.35 Total number of infant HM pauses (side view
camera) during free interaction Video, infant 0.34 Total number of infant HM
pauses (side view camera) when the mother is singing Video, infant 0.32 Vertical
amplitude of the giraffe (front view camera) Video, infant -0.30 Movement
acceleration max (side view camera) during free interaction Video, infant 0.29
West with ASD/ID vs. West without ASD/ID Total number of all infant vocalization
during free interaction Audio, infant -0.56 Synchrony ratio (infant response to
mother) Audio, synchrony -0.55 Ratio of all infant vocalization during free
interaction Audio, infant -0.55 Motherese synchrony ratio (infant response to
motherese) Audio, synchrony -0.54 Non-motherese synchrony ratio (infant response
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during the giraffe interaction Video, infant -0.46 HM acceleration max (side
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(front view camera) during the giraffe interaction Video, infant -0.43 Curvature
max (side view camera) during the giraffe interaction Video, infant -0.37
Relative time spent motionless (pause) (front view camera) during free
interaction Video, infant 0.36 p-value 0.012 0.014 0.023 0.032 0.034 <0.001
<0.001 0.001 0.002 0.005 0.008 0.01 0.014 0.039 0.04HM hand movement, ASD autism
spectrum disorder, ID intellectual disability, SD standard deviation.',
 'sentences': ['2| o Confusion Machine learning classification Decision stump
(leave one out) matrices I West vs. TD (n=51) 90 Clas: ied West T 80 as D West
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with West with 20 ID/ASD out ID/ASD West with 5 5 10 ID/ASD West with 1 21
Mutimodal Video Audio out ID/ASDFig.',
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'page_sentence_count_spacy': 3,

'sentence_chunks': [['2| o Confusion Machine learning classification Decision stump (leave one out) matrices I West vs. TD (n=51) 90 Clas: ied West T 80 as D West 32 70 el 12 60 50 West with ID/ASD vs. 40 I West with out ID/ASD 30 (N=32) West with West with 20 ID/ASD out ID/ASD West with 5 5 10 ID/ASD West with 1 21 Mutimodal Video Audio out ID/ASDFig.'],

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'num_chunks': 1},

{'page_number': 6,

'text': 'synchrony vocalization features (N = 3) and infant HM features (N = 5), the last of which showed lower correla- tion scores.DiscussionTo the best of our knowledge, this is the rst study to apply multimodal to mother-infant

interactions in the context of WS. Combining an infant-mother interaction at 9 months significantly predicted the development of ASD or severe to moderate ID at 4 years of age in the high-risk children with WS. Confusion matrices showed that the classification errors were not random, enhancing the interest of the computational method proposed here. In addition, the best contributing features for the performed classifications differed when classifying WS vs. TD and WS+ vs. WS-. Infant HMs were the most significant features to distinguish WS versus TD, probably reflecting the motor impact due to acute WS encephalopathy. For classifying WS+ vs. WS-, the contribution of infant audio features and synchrony features became much more relevant combined with several HM features. We believe that the importance of synchrony and reciprocity during early interactions is in line with recent studies that have investigated the risk of ASD or NDD during the first year of life from home movies (e.g., refs. 11,24), from prospective follow-up of high-risk infants such as siblings (e.g., refs. 4,28) or infants with WS (e.g., ref. 14), and from prospective studies assessing tools to screen risk for autism (e.g., ref. 29). In the field of ASD, synchrony, reciprocity, parental sensitivity, and emotional engagement are now proposed as targets of early interventions³⁰, which could prevent early interactive vicious circles. Parents of at-risk infants try to compensate for the lack of interactivity of their child by modifying their stimulation and therefore sometimes interactions²⁴. Early reinforcing identification of these interactive targets is especially useful among babies with neurological comorbidities because delays and impairments in early social interactions are not sufficient to predict ASD. Similarly, we believe that the importance of HM in distinguishing WS vs. TD on one hand, and WS+ vs. WS- on the other hand, is also in line with the studies that investigated the importance of non-social behaviors for investigating the risk of ASD or NDD during the first year of life. For example, studying home movies, Purpura et al. found more bilateral HM and finger movements in infants who will later develop ASD³¹. Similarly, several prospective follow-up studies of high-risk siblings³²⁻³⁵ or retrospective studies on home movies^{36,37} reported specific motor atypical repertoire in infants with ASD. In ASD, early social signals have previously been assessed with automatized and computational procedures, focusing on eye tracking at early stages³⁸⁻⁴⁰, vocal productions⁴¹, analysis of acoustics of first utterances or cry episodes⁴², but none was done in an interactive setting. Our study proposed a paradigm shift from the assessment of infant behavior to dyadic assessment of interactions, as previously achieved in retrospective approaches using home movies²⁴. The aim is not to implement studies of social signal processing in routine clinical work but rather to decompose clinical intuitions and signs and validate the most relevant cues of these clinical features. From clinical work, back to clinics, social signal processing is a rigorous step to help clinicians better identify and assess early targets of interventions. Given the exploratory nature of both our approach and method, our results should be interpreted with caution taking into account strengths (prospective follow-up, automatized multimodal social signal processing, and ecological standardized assessment) and limitations. These limitations include (1) the overall sample size knowing that WS is a rare disease; (2) the high rate of missing data during video recording

due to the ecological conditions of the infant-mother interaction (mothers interposing between the camera and the infant); the small sample size of WS+ (N = 10) that limited the power of machine learning methods. We conclude that the method proposed here combining multimodal automated assessment of social signal processing during early interaction with infants at risk for NDD is a promising tool to decipher clinical features that remain difficult to identify and assess. In the context of WS, we showed that such a method we proposed to label 'behavioral and interaction imaging' was able to significantly predict the development of ASD or ID at 4 years of age in high-risk children who had WS and were assessed at 9 months of age. Acknowledgements The authors thank all of the patients and families who participated in this study. The study was funded by the EADS foundation (PILE), by the Agence Nationale de la Recherche (ANR-12-SAMA-006-1) and the Groupement de Recherche en Psychiatrie (GDR-3557). It was partially performed in the Labex SMART (ANR-11-LABX-65), which is supported by French state funds and managed by the ANR in the Investissements d'Avenir program under reference ANR-11-IDEX-0004-02. The sponsors had no involvement in the study design, data analysis, or interpretation of the results. Author details 1Service de Psychiatrie de l'Enfant, AP-HP, Hôpital Necker, 149 rue de Sèvres, 75015 Paris, France. 2Institut des Systèmes Intelligents et de Robotique, CNRS, UMR 7222, Sorbonne Université, 4 Place Jussieu, 75252 Paris Cedex, France. 3Département de Psychiatrie de l'Enfant et de l'Adolescent, AP-HP, Hôpital Pitié-Salpêtrière, 47-83, Boulevard de l'Hôpital, 75651 Paris, Cedex 13, France. 4Ariana Pharmaceuticals, Research Department, Paris, France. 5Service de Neuropédiatrie, AP-HP, Hôpital Necker, 136, Rue de Vaugirard, 75015 Paris, France',

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'page_sentence_count_spacy': 33,

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'num_chunks': 4},

{'page_number': 7,

'text': 'Conflict of interest The authors declare that they have no conflict of interest. Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations. Supplementary Information accompanies this paper at

(<https://doi.org/10.1038/s41398-020-0743-8>). Received: 7 December 2019 Revised: 13 January 2020 Accepted: 16 January 2020 Published online: 03 February 2020',

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Accepted: 16 January 2020Published online: 03 February 2020']],
'num_chunks': 1}}}

0.0.3 Splitting each chunk into its own item

```
[23]: import re
# Split each chunk into its own item
pages_and_chunks = []
for pdf in loaded_pdf_and_text:
    for page in pdf['text']:
        for sentence_chunk in page["sentence_chunks"]:
            chunk_dict = {}
            chunk_dict["pdf_name"] = pdf['filename']
            chunk_dict["page_number"] = page["page_number"]

            # Join the sentences together into a paragraph-like structure, aka a
            ↪ chunk (so they are a single string)
            joined_sentence_chunk = "".join(sentence_chunk).replace(" ", " ").
            ↪ strip()
            joined_sentence_chunk = re.sub(r'\.([A-Z])', r'.
            ↪ \1', joined_sentence_chunk)
            chunk_dict["sentence_chunk"] = joined_sentence_chunk
            # Get stats about the chunk
            chunk_dict["chunk_char_count"] = len(joined_sentence_chunk)
            chunk_dict["chunk_word_count"] = len([word for word in
            ↪ joined_sentence_chunk.split(" ")])
            chunk_dict["chunk_token_count"] = len(joined_sentence_chunk) / 4 # 1
            ↪ token = ~4 characters
            pages_and_chunks.append(chunk_dict)

# How many chunks do we have?
len(pages_and_chunks)
```

[23]: 517

```
[24]: #Let's print some random sample
import random
random.sample(pages_and_chunks, k=1)
```

```
[24]: [{'pdf_name': 'Patten_Audio.pdf',
      'page_number': 1,
      'sentence_chunk': '"', NAUo% 3 P HENPublished in final edited form as:J Autism
Dev Disord.2014 October ; 44(10): 2413-2428.doi:10.1007/s10803-014-2047-4. Vocal
patterns in infants with Autism Spectrum Disorder: Canonical babbling status and
vocalization frequencyElena Patten, Ph. D.1, Katie Belardi, M. S.2, Grace T.
Baranek, Ph. D.2, Linda R. Watson, Ed. D. 2, Jeffrey D. Labban, Ph. D.1, and D.
```

```

Kimbrough Oller, Ph. D.3 1Univ.of North Carolina, Greensboro2Univ.of North
Carolina, Chapel Hill3Univ.',
    'chunk_char_count': 480,
    'chunk_word_count': 68,
    'chunk_token_count': 120.0}]

```

```

[25]: # Get stats about our chunks
import pandas as pd
df = pd.DataFrame(pages_and_chunks)
df.describe().round(2)

```

```

[25]:      page_number  chunk_char_count  chunk_word_count  chunk_token_count
count          517.00           517.00           517.00           517.00
mean             9.48          1187.22           182.33           296.80
std              7.16           710.37           114.55           177.59
min              1.00             5.00             1.00             1.25
25%              4.00           546.00             77.00           136.50
50%              7.00          1197.00           184.00           299.25
75%             16.00          1666.00           256.00           416.50
max             35.00          4075.00           647.00          1018.75

```

```

[26]: # Show random chunks with under 30 tokens in length
min_token_length = 30
for row in df[df["chunk_token_count"] <= min_token_length].sample(5).iterrows():
    print(f'Chunk token count: {row[1]["chunk_token_count"]} | Text:
    ↳{row[1]["sentence_chunk"]}')

```

Chunk token count: 3.0 | Text:Through this

Chunk token count: 13.0 | Text:Author manuscript; available in PMC 2014 October 01.

Chunk token count: 28.75 | Text:What did the researchers do and find?PLOS Medicine | <https://doi.org/10.1371/journal.pmed.1002705> November 27, 2018

Chunk token count: 7.5 | Text:Automatic diagnostic procedure

Chunk token count: 29.25 | Text:For the above, despite the novelty@ SpringerContent courtesy of Springer Nature, terms of use apply. Rights reserved.

0.0.4 Let's filter our DataFrame/list of dictionaries to only include chunks with over 30 tokens in length.

```

[27]: pages_and_chunks_over_min_token_len = df[df["chunk_token_count"] >=
    ↳min_token_length].to_dict(orient="records")
pages_and_chunks_over_min_token_len[:2]

```

```

[27]: [{'pdf_name': 'LEE.pdf',
      'page_number': 1,
      'sentence_chunk': '.sensorsbyLetterLetter Deep-Learning-Based Detection of

```

Infants with Autism Spectrum Disorder Using Auto-Encoder Feature

RepresentationJung Hyuk Lee 1, Geon Woo Lee 1, Guiyoung Bong 2, Hee Jeong Yoo 2,3 and Hong Kook Kim 1,*1School of Electrical Engineering and Computer Science, Gwangju Institute of Science and Technology, Gwangju 61005, Korea;

ljh0412@gist.ac.kr (J. H. L.); geonwoo0801@gist.ac.kr (G. W. L.) 2 Department of Psychiatry, Seoul National University Bundang Hospital, Seongnam-si, Gyeonggi-do 13620, Korea; 20409@snu.ac.kr (G. B.); hjyoo@snu.ac.kr (H. J. Y.) 3 Department of Psychiatry, College of Medicine, Seoul National University, Seoul 03980, Korea * Correspondence: hongkook@gist.ac.krReceived: 29 October 2020; Accepted: 24

November 2020; Published: 26 November 2020check for v updatesAbstract: Autism spectrum disorder (ASD) is a developmental disorder with a life-span disability. While diagnostic instruments have been developed and qualified based on the accuracy of the discrimination of children with ASD from typical development (TD) children, the stability of such procedures can be disrupted by limitations pertaining to time expenses and the subjectivity of clinicians. Consequently, automated diagnostic methods have been developed for acquiring objective measures of autism, and in various fields of research, vocal characteristics have not only been reported as distinctive characteristics by clinicians, but have also shown promising performance in several studies utilizing deep learning models based on the automated discrimination of children with ASD from children with TD. However, difficulties still exist in terms of the characteristics of the data, the complexity of the analysis, and the lack of arranged data caused by the low accessibility for diagnosis and the need to secure anonymity. In order to address these issues, we introduce a pre-trained feature extraction auto-encoder model and a joint optimization scheme, which can achieve robustness for widely distributed and untrained data using a deep-learning-based method for the detection of autism that utilizes various models. By adopting this auto-encoder-based feature extraction and joint optimization in the extended version of the Geneva minimalistic acoustic parameter set (eGeMAPS) speech feature data set, we acquire improved performance in the detection of ASD in infants compared to the raw data set. Keywords: auto-encoder; bidirectional long short-term memory (BLSTM); joint optimization; acoustic feature extraction; autism spectrum disorder1. IntroductionAutism spectrum disorder (ASD) is a developmental disorder with a high probability of causing difficulties in social interactions with other people [1]. According to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), ASD involves several characteristics such as being con ned to specific interests or behaviors, delayed linguistic development, and poor functionality in terms of communicating or functioning in social situations [2].',

```
'chunk_char_count': 3028,
'chunk_word_count': 414,
'chunk_token_count': 757.0},
{'pdf_name': 'LEE.pdf',
'page_number': 1,
```

```
'sentence_chunk': 'As there is wide variation in terms of the types and
severities of ASD based on its characteristics, the disorder is referred to as a
spectrum [1]. Not only does ASD have the characteristics of a developmental
```

disorder with a life-span disability, but its prevalence is also increasing—from 1 in 150 children in 2000 to 1 in 54 children in 2016 [3]. As diverse evidence has been obtained from previous research showing that the chance of improvement in theSensors 2020, 20, 6762;

```
doi:10.3390/s20236762www.mdpi.com/journal/sensors',
  'chunk_char_count': 529,
  'chunk_word_count': 83,
  'chunk_token_count': 132.25}]
```

1 Embedding our text chunks

```
[28]: from sentence_transformers import SentenceTransformer
      embedding_model = SentenceTransformer(model_name_or_path="all-mpnet-base-v2")
```

```
modules.json: 0%|          | 0.00/349 [00:00<?, ?B/s]
config_sentence_transformers.json: 0%|          | 0.00/116 [00:00<?, ?B/s]
README.md: 0%|          | 0.00/10.6k [00:00<?, ?B/s]
sentence_bert_config.json: 0%|          | 0.00/53.0 [00:00<?, ?B/s]
config.json: 0%|          | 0.00/571 [00:00<?, ?B/s]
model.safetensors: 0%|          | 0.00/438M [00:00<?, ?B/s]
tokenizer_config.json: 0%|          | 0.00/363 [00:00<?, ?B/s]
vocab.txt: 0%|          | 0.00/232k [00:00<?, ?B/s]
tokenizer.json: 0%|          | 0.00/466k [00:00<?, ?B/s]
special_tokens_map.json: 0%|          | 0.00/239 [00:00<?, ?B/s]
1_Pooling/config.json: 0%|          | 0.00/190 [00:00<?, ?B/s]
```

```
[29]: import time
```

```
[30]: # Create embeddings one by one
      for item in pages_and_chunks_over_min_token_len:
          item["embedding"] = embedding_model.encode(item["sentence_chunk"])
```

```
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
Batches: 0%|          | 0/1 [00:00<?, ?it/s]
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```

Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]
Batches:  0%|          | 0/1 [00:00<?, ?it/s]

```

2 We can perform batched operations by turning our target text samples into a single list and then passing that list to our embedding model.

```

[31]: # Turn text chunks into a single list
text_chunks = [item["sentence_chunk"] for item in
               ↪pages_and_chunks_over_min_token_len]

```

```

[32]: %%time
# Embed all texts in batches
text_chunk_embeddings = embedding_model.encode(text_chunks,
        ↪batch_size=32,convert_to_tensor=True)
text_chunk_embeddings

```

```

Batches:  0%|          | 0/16 [00:00<?, ?it/s]

```

```

CPU times: user 9min 24s, sys: 2min 49s, total: 12min 14s
Wall time: 6min 10s

```

```
[32]: tensor([[[-0.0206,  0.0329, -0.0465, ..., -0.0200, -0.0548, -0.0453],
               [-0.0653, -0.0180, -0.0537, ..., -0.0223, -0.0255, -0.0312],
               [-0.0456,  0.0325, -0.0454, ...,  0.0114, -0.0225, -0.0467],
               ...,
               [ 0.0057, -0.0318, -0.0421, ..., -0.0134, -0.0218, -0.0245],
               [ 0.0036, -0.0277, -0.0399, ..., -0.0138, -0.0211, -0.0223],
               [-0.0039, -0.0518, -0.0254, ..., -0.0199, -0.0702, -0.0446]])
```

3 Save embeddings to file

3.0.1 Since creating embeddings can be a timely process, let's turn our `pages_and_chunks_over_min_token_len` list of dictionaries into a DataFrame and save it.

```
[33]: # Save embeddings to file
text_chunks_and_embeddings_df = pd.
↳ DataFrame(pages_and_chunks_over_min_token_len)
embeddings_df_save_path = "/kaggle/working/text_chunks_and_embeddings_df.csv"
text_chunks_and_embeddings_df.to_csv(embeddings_df_save_path, index=False)
```

```
[35]: # Import saved file and view
import pandas as pd
embeddings_df_save_path = "/kaggle/working/text_chunks_and_embeddings_df.csv"
text_chunks_and_embedding_df_load = pd.read_csv(embeddings_df_save_path)
text_chunks_and_embedding_df_load.head()
```

```
[35]: pdf_name  page_number  sentence_chunk \
0  LEE.pdf          1  .sensorsbyLetterLetter Deep-Learning-Based Det...
1  LEE.pdf          1  As there is wide variation in terms of the typ...
2  LEE.pdf          2  social abilities of people with ASD increases ...
3  LEE.pdf          2  For examples of machine learning classifi cation...
4  LEE.pdf          3  di culties pertaining to the need to secure th...
```

```
chunk_char_count  chunk_word_count  chunk_token_count \
0              3028              414              757.00
1               529               83              132.25
2             3061             450             765.25
3             1790             255             447.50
4             1718             264             429.50
```

```
embedding
0  [-2.05749348e-02  3.28779072e-02 -4.64697927e-...
1  [-6.53321519e-02 -1.79831404e-02 -5.37393503e-...
2  [-4.55831960e-02  3.25354822e-02 -4.53654565e-...
3  [-4.83638458e-02  3.05312667e-02 -3.35919224e-...
4  [-3.53161991e-02  3.77684757e-02 -4.86659035e-...
```

4 Similarity search

4.0.1 Let's import our embeddings we created earlier (tk-link to embedding file) and prepare them for use by turning them into a tensor.

```
[36]: import random
import torch
import numpy as np
import pandas as pd

# Import texts and embedding df
text_chunks_and_embedding_df = pd.read_csv("/kaggle/working/
↳text_chunks_and_embeddings_df.csv")
# Convert embedding column back to np.array (it got converted to string when it
↳got saved to CSV)
text_chunks_and_embedding_df["embedding"] =
↳text_chunks_and_embedding_df["embedding"].apply(lambda x: np.fromstring(x.
↳strip("[]"), sep=" "))
# Convert texts and embedding df to list of dicts
pages_and_chunks = text_chunks_and_embedding_df.to_dict(orient="records")
# # Convert embeddings to torch tensor and send to device (note: NumPy arrays
↳are float64, torch tensors are float32 by default)
embeddings = torch.tensor(np.array(text_chunks_and_embedding_df["embedding"].
↳tolist()), dtype=torch.float32)
embeddings.shape
```

```
[36]: torch.Size([503, 768])
```

```
[37]: text_chunks_and_embedding_df.head()
```

```
[37]: pdf_name  page_number  sentence_chunk \
0  LEE.pdf      1  .sensorsbyLetterLetter Deep-Learning-Based Det...
1  LEE.pdf      1  As there is wide variation in terms of the typ...
2  LEE.pdf      2  social abilities of people with ASD increases ...
3  LEE.pdf      2  For examples of machine learning classi cation...
4  LEE.pdf      3  di culties pertaining to the need to secure th...

chunk_char_count  chunk_word_count  chunk_token_count \
0              3028              414              757.00
1              529              83              132.25
2             3061             450              765.25
3             1790             255              447.50
4             1718             264              429.50

embedding
0  [-0.0205749348, 0.0328779072, -0.0464697927, -...
1  [-0.0653321519, -0.0179831404, -0.0537393503, ...
```

```

2 [-0.045583196, 0.0325354822, -0.0453654565, 0...
3 [-0.0483638458, 0.0305312667, -0.0335919224, 0...
4 [-0.0353161991, 0.0377684757, -0.0486659035, 0...

```

```
[38]: embeddings[0]
```

```

[38]: tensor([-2.0575e-02,  3.2878e-02, -4.6470e-02, -9.3290e-03,  2.2779e-02,
              2.9414e-02,  4.6374e-02, -3.1967e-02, -6.6183e-02, -1.5472e-02,
              6.5602e-03, -5.9365e-02,  6.3316e-02,  6.2994e-02, -5.6251e-02,
              -2.7083e-02, -8.2688e-03,  3.3644e-03,  3.7265e-02, -8.6292e-03,
              3.4178e-02,  1.0568e-02, -6.3263e-02,  7.3406e-02, -4.4935e-02,
              3.0851e-02, -3.3910e-03, -6.0136e-02,  7.7574e-03, -6.9380e-02,
              -1.0476e-03,  8.3198e-03,  2.3188e-02,  6.4961e-02,  2.1899e-06,
              -2.7092e-02,  1.1488e-02, -2.1731e-02, -3.3694e-02, -6.7440e-02,
              -3.4016e-02, -2.0057e-02,  2.5766e-03,  3.3616e-02, -3.1582e-02,
              -8.3078e-02,  3.9643e-02,  2.4284e-02,  2.3357e-03,  2.6866e-02,
              -1.3909e-02,  1.7111e-02,  6.4790e-02,  1.7834e-02,  1.4242e-02,
              -1.0971e-02,  7.7212e-03, -4.0157e-02,  1.0227e-02, -4.5842e-02,
              -4.0227e-02,  3.4358e-02, -9.8436e-03,  1.1578e-02,  2.8985e-02,
              -1.0710e-02, -5.1550e-03, -7.0833e-02, -7.9531e-03,  5.5576e-02,
              6.5233e-02,  3.6808e-02,  1.2630e-02,  2.1378e-02, -2.5656e-02,
              -2.2481e-02, -2.5487e-02,  1.3879e-02,  2.4961e-04,  2.3939e-02,
              -1.6155e-02,  1.0639e-02, -8.1041e-03, -6.0489e-02,  2.9155e-02,
              7.7753e-03, -2.7845e-02,  2.6396e-02, -3.0354e-03, -2.1695e-03,
              2.0621e-02, -7.4493e-03,  3.9614e-02,  2.2232e-02, -5.1900e-02,
              -1.2570e-02, -6.8271e-02, -5.8859e-02, -8.3609e-03,  4.6870e-02,
              -9.7581e-03, -2.1877e-02, -6.7945e-02, -2.5251e-02, -2.9048e-02,
              5.5710e-02,  4.0973e-03, -1.0147e-03, -3.6789e-02,  1.3450e-02,
              8.0660e-04,  2.3622e-02,  1.7422e-03,  2.7656e-03,  7.1559e-02,
              -2.8429e-02, -7.7080e-02, -8.7857e-03, -9.7261e-03, -1.1967e-02,
              -1.4766e-04,  4.8303e-02, -2.3753e-02,  6.4569e-02,  1.5716e-02,
              2.1628e-02,  7.0274e-03,  2.6115e-02,  1.8574e-03, -2.6593e-02,
              -1.0740e-02, -2.6220e-03,  5.2512e-02,  4.1975e-02,  5.8415e-04,
              1.8094e-02,  8.2413e-02,  6.0421e-03,  3.3156e-02,  3.1979e-02,
              -2.8922e-02,  1.1672e-02,  3.5141e-02, -2.5643e-02,  3.0980e-04,
              3.6174e-02,  7.2161e-04, -2.9451e-02,  3.1448e-03, -3.9861e-02,
              8.0228e-02, -1.9446e-02, -1.1915e-02,  2.1233e-03,  2.3937e-02,
              -2.7930e-02,  3.6772e-02,  4.1132e-02,  5.5125e-02, -6.9711e-02,
              2.1080e-02, -6.3639e-02, -1.8086e-02,  1.7592e-02, -2.6750e-02,
              -3.6126e-02, -2.0252e-03, -6.9899e-02, -4.0183e-02,  4.8303e-02,
              -2.6887e-03,  3.6741e-03,  4.0855e-02, -8.1944e-03, -2.5518e-05,
              7.4415e-02,  3.3829e-02, -6.4120e-02, -2.0504e-02,  3.2299e-03,
              4.4087e-03, -3.1259e-02,  5.0820e-02,  3.2110e-02, -1.9663e-02,
              -4.3919e-02,  2.7564e-03,  2.7715e-02, -7.8673e-03, -3.7913e-02,
              -4.0924e-02, -5.5524e-02, -2.7134e-02,  7.0631e-02,  5.0223e-02,
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```

-1.5957e-02, 4.6858e-03, -5.6100e-02, -4.4937e-02, -4.1214e-02,
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 2.3136e-02, -2.1816e-02, 1.2926e-02, 7.5811e-02, -7.8506e-02,
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 2.2264e-02, -3.6317e-02, -9.0572e-03, 4.3532e-02, -3.8315e-02,
 -5.0513e-02, -6.8330e-02, 1.2168e-02, -2.6492e-02, 6.8981e-02,
 -4.5591e-02, 5.8373e-03, 1.5929e-02, -3.9251e-02, -3.3305e-02,
 6.0449e-02, -4.9306e-03, -2.9464e-04, 9.5247e-03, 3.0050e-02,
 -6.9178e-02, -1.7409e-02, 1.4804e-02, -2.7513e-02, 5.8696e-02,
 -7.0098e-02, -1.2958e-02, 7.0048e-03, -6.5640e-02, 2.2013e-02,
 8.2572e-04, 6.2169e-02, 2.0745e-02, -2.2125e-02, -2.3158e-02,
 -1.1270e-02, 1.9324e-02, 6.9733e-02, -3.8143e-02, -4.8050e-02,
 2.8382e-02, 1.9725e-02, -2.7919e-02, 2.8754e-02, -8.9622e-03,
 -7.5089e-02, 2.0756e-03, 1.2414e-01, -2.0108e-02, 8.1890e-02,
 -4.3505e-03, 1.7116e-02, -7.1590e-03, 3.3729e-02, 2.6662e-02,
 1.9097e-02, -1.8581e-04, 7.9112e-04, -4.2909e-03, -1.2594e-02,
 1.8923e-02, 3.1145e-02, 1.4269e-02, -1.0579e-02, -3.1407e-02,
 -1.8561e-02, -4.1905e-02, -1.0694e-02, 6.4689e-03, 2.8228e-02,
 8.4760e-03, -2.6760e-02, 2.4092e-03, -6.9729e-02, 2.3121e-02,
 4.5761e-02, -9.5333e-03, 2.4443e-02, 5.9941e-04, -6.1767e-04,
 -1.7765e-02, 2.6409e-03, -2.8392e-02, -5.8725e-04, -1.5095e-02,
 2.4023e-02, 3.7996e-02, -3.6332e-02, -2.3345e-02, 2.6283e-02,
 2.5858e-02, -4.3495e-02, 1.9747e-02, -4.2185e-02, 1.1453e-02,
 1.5553e-02, -1.4669e-02, -8.1585e-03, -1.9433e-02, -4.3742e-02,
 6.5356e-03, -1.2682e-02, 1.7927e-03, -2.7383e-02, -4.7899e-03,
 -5.5257e-33, -6.6879e-03, -1.7007e-02, -1.1973e-02, 1.1705e-02,
 -2.5753e-02, 5.5998e-02, 1.2167e-02, 5.2910e-02, -3.5370e-02,
 -2.5710e-02, -2.6362e-03, -1.9915e-02, 2.2006e-02, -4.5409e-03,
 3.0181e-02, -2.2691e-02, 2.2855e-02, -3.5890e-02, 2.1701e-02,
 -1.3312e-03, 5.4414e-03, 3.9403e-03, 8.0106e-02, -7.6395e-02,
 -2.8341e-02, 5.3958e-02, 1.2376e-02, -4.8538e-02, 3.7952e-02,
 6.8424e-04, -1.0652e-02, 1.3664e-02, 3.8249e-04, 5.4214e-02,
 6.9714e-03, 9.0640e-03, -6.7273e-02, -2.5609e-03, -7.0502e-02,
 3.8416e-02, -6.3117e-02, 1.6781e-02, 7.5201e-02, -1.1479e-02,
 -3.3353e-02, -1.0558e-02, 4.9580e-02, -3.2960e-02, 1.4573e-02,
 -2.6200e-02, -2.4146e-03, -5.9786e-03, -1.2046e-02, -1.5075e-02,
 5.1204e-02, 4.2488e-02, -5.1668e-02, -1.1917e-03, -3.9055e-02,
 1.8226e-02, 4.4821e-02, 1.7298e-02, 9.6132e-03, 8.5234e-03,
 6.7495e-02, 2.5469e-02, 6.1250e-02, -3.5350e-02, -5.2576e-03,
 5.9310e-03, 5.7702e-02, 7.1102e-03, 3.0668e-02, -1.0360e-02,
 1.9408e-02, -2.5786e-02, -7.2911e-02, 4.8559e-02, -1.6762e-02,
 3.9819e-02, 3.3233e-03, -7.6013e-03, -7.9810e-02, 2.8594e-02,
 -1.8562e-02, 4.0246e-02, -1.6942e-02, -2.8469e-02, 1.7323e-02,
 -2.3825e-02, -5.0965e-02, -1.0849e-02, 1.9295e-02, -6.5963e-03,
 2.2670e-02, -3.5517e-02, 1.1191e-01, 5.3301e-03, 3.5446e-02,
 8.9107e-03, -1.4356e-03, -5.4501e-02, 5.6265e-02, 1.3525e-03,
 6.9223e-02, -1.5568e-02, -2.8458e-02, 1.6539e-02, -7.7397e-03,
 -2.3679e-02, 3.4587e-02, 8.3195e-04, 2.2314e-02, 1.8731e-02,
 -1.5854e-02, -2.0625e-02, -3.9120e-03, 1.1938e-02, -8.7973e-03,

```
-5.4329e-04, -9.5267e-02, 1.2402e-05, 5.9279e-03, 8.7424e-03,
-1.9634e-02, -2.2033e-02, -5.4507e-02, -5.5146e-02, 3.3753e-02,
-3.4869e-02, 2.5878e-02, -4.4991e-02, 2.8291e-07, -5.8799e-02,
8.2761e-03, 1.7987e-02, -2.8159e-02, 3.0217e-02, 2.7607e-02,
-1.7815e-02, 1.3617e-02, 3.9300e-02, -2.1017e-02, 2.7954e-02,
3.3207e-02, 2.0454e-02, 3.2351e-02, -1.0821e-02, -1.7495e-03,
-1.1889e-02, 6.7653e-02, -7.3867e-03, 3.9430e-02, -1.2415e-02,
5.4017e-02, -1.2774e-02, -5.4741e-02, 5.3151e-02, 3.0406e-04,
-9.6145e-03, 3.4701e-02, 4.0834e-02, -2.0143e-02, -1.5076e-03,
-4.1403e-02, 7.2810e-02, 7.6652e-03, -2.5012e-02, 4.6883e-02,
-3.7657e-02, -2.4387e-02, -6.7411e-02, -5.2313e-02, 5.0960e-03,
7.9739e-02, -6.6768e-03, 1.8942e-03, 9.3179e-02, -2.5475e-02,
-2.2310e-02, -6.5265e-02, 9.8046e-03, -2.7648e-02, 7.4665e-02,
-1.8539e-02, 3.8185e-02, 3.9247e-03, 1.2617e-03, -3.6228e-02,
-1.6230e-02, -1.8062e-02, 8.7821e-03, 9.1345e-03, -5.4996e-02,
-1.7317e-02, 2.5722e-02, 3.6936e-02, -5.0455e-02, 2.8000e-02,
-6.1680e-03, 2.1387e-34, -1.0410e-02, -1.7201e-02, -3.1504e-04,
-4.8171e-02, 1.7879e-03, -4.3859e-03, -7.4677e-03, 1.2815e-02,
-1.9996e-02, -5.4800e-02, -4.5315e-02])
```

4.1 Now let's prepare another instance of our embedding model. Not because we have to but because we'd like to make it so you can start the notebook from the cell above.

```
[39]: from sentence_transformers import util, SentenceTransformer
embedding_model = SentenceTransformer(model_name_or_path="all-mpnet-base-v2")
```

```
[40]: # 1. Define the query
# Note: This could be anything. But since we're working with a specific set of
#       ↪pdfs, we'll stick with queries related to those pdfs.
query = "Autism spectrum disorder"
print(f"Query: {query}")
# 2. Embed the query to the same numerical space as the text examples
# Note: It's important to embed your query with the same model you embedded
#       ↪your examples with.
query_embedding = embedding_model.encode(query, convert_to_tensor=True)
# 3. Get similarity scores with the dot product (we'll time this for fun)
from time import perf_counter as timer
start_time = timer()
dot_scores = util.dot_score(a=query_embedding, b=embeddings)[0]
end_time = timer()

print(f"Time take to get scores on {len(embeddings)} embeddings:
#       ↪{end_time-start_time:.5f} seconds.")

# 4. Get the top-k results (we'll keep this to 5)
```

```
top_results_dot_product = torch.topk(dot_scores, k=5)
top_results_dot_product
```

Query: Autism spectrum disorder

Batches: 0%| | 0/1 [00:00<?, ?it/s]

Time take to get scores on 503 embeddings: 0.00051 seconds.

```
[40]: torch.return_types.topk(
      values=tensor([0.6785, 0.6635, 0.6534, 0.6270, 0.6262]),
      indices=tensor([394, 497, 390, 55, 232]))
```

4.2 Let's check the results of our similarity search.

First, we'll define a small helper function to print out wrapped text (so it doesn't print a whole text chunk as a single line).

```
[41]: # Define helper function to print wrapped text
import textwrap
def print_wrapped(text, wrap_length=80):
    wrapped_text = textwrap.fill(text, wrap_length)
    print(wrapped_text)
```

```
[42]: print(f"Query: '{query}'\n")
      print("Results:")
      # Loop through zipped together scores and indicies from torch.topk
      for score, idx in zip(top_results_dot_product[0], top_results_dot_product[1]):
          print(f"Score: {score:.4f}")
          # Print relevant sentence chunk (since the scores are in descending order, the
          ↪ most relevant chunk will be first)
          print("Text:")
          print_wrapped(pages_and_chunks[idx]["sentence_chunk"])
          # Print the page number too so we can reference the textbook further (and
          ↪ check the results)
          print(f"Page number: {pages_and_chunks[idx]['page_number']}")
          print("\n")
```

Query: 'Autism spectrum disorder'

Results:

Score: 0.6785

Text:

PMCID: PMC3424065. Lord C, Rutter M, Goode S, Heemsbergen J, Jordan H, Mawhood L, et al. Autism diagnostic observation schedule: A standardized observation of communicative and social behavior. Journal of autism and developmental disorders. 1989; 19(2):185-212. PMID: 2745388. Lord C, Rutter M, Le Couteur A. Autism Diagnostic Interview-Revised: a revised version of a diagnostic interview for caregivers of individuals with possible pervasive developmental disorders.

Journal of autism and developmental disorders.1994; 24(5):659-85. PMID: 7814313.10. Association AP. Diagnostic and statistical manual of mental disorders (DSM-5®).

Page number: 18

Score: 0.6635

Text:

Patten et al. Page 30Table 1Participant DemographicsASD; n=23 TD; n=14 Age at 9-12months; mean (SD) 10.89 (1.39) 10.63 (.53) Age at 15-18 months; mean (SD) 16.33 (.83) 16.28 (.70) Sex 19 males, 4 females 11 males, 3 females Race 23 White, 1 Black 13 White, 1 Asian Maternal education 1 5.48 5.82 Childhood Autism Rating Scale; mean (SD) 34.17 (1.52) 16.15 (.39)31Maternal education: 1=6th grade or lower; 2=7th to 9th grade; 3=partial high school; 4=high school graduate/GED; 5=associate of arts/associate of science or technical training or partial college training; 6=bachelor of arts/science; 7=master of arts/science or doctorate or other professional degree completed2missing information for two participants3missing information for four participantsJ Autism Dev Disord. Author manuscript; available in PMC 2014 October 01.

Page number: 30

Score: 0.6534

Text:

@PLOS | MEDICINEReferences 1. Prince M, Patel V, Saxena S, Maj M, Maselko J, Phillips MR, et al. Global mental health 1 - No health without mental health. Lancet.2007; 370(9590):859-77.[https://doi.org/10.1016/S0140-6736\(07\)61238-0](https://doi.org/10.1016/S0140-6736(07)61238-0) PMID: 178040632. Baio J, Wiggins L, Christensen DL, Maenner MJ, Daniels J, Warren Z, et al. Prevalence of Autism Spec- trum Disorder Among Children Aged 8 Years-Autism and Developmental Disabilities Monitoring Net- work, 11 Sites, United States, 2014. MMWR Surveillance Summaries.2018; 67(6):1.

Page number: 18

Score: 0.6270

Text:

TABLE 1 | Comparison of the general conditions between the HR group and the TD group (mean $\pm$ SD). HR group (n = 45) TD group (n = 43) t/2 P-value Sex -0.06 0.08 Male 40 32 Age (months) DQ Female 5 19.71 $\pm$ 3.43 11 16.40 $\pm$ 4.70 3.80 <0.01 92.98 $\pm$ 7.89 -4.34 <0.01 92.77 $\pm$ 8.46 -0.25 93.70 $\pm$ 8.29 -2.24 80.29 $\pm$ 17.62 92.02 $\pm$ 17.60 86.64 $\pm$ 19.03 0.03 60.84 $\pm$ 21.27 86.51 $\pm$ 8.353 -7.39 <0.01 92.28 $\pm$ 7.18 -4.76 <0.01 Adaptive Gross motor 0.80 Fine motor Language Personal-social 78.80 $\pm$ 17.19HR, high-risk autism spectrum disorder; TD, typical development; DQ, developmental quotient of the Gesell Developmental Scale. Diagnostic Evaluation of Children in the HR Group at 2 Years of Age At 2 years of age, the children in the HR group were evaluated using the Autism Diagnosis Interview-R (ADI-R) and the Autism Diagnostic Observation Schedule (ADOS). Two pediatric psychiatrists then clinically diagnosed the children based on the ASD diagnostic

criteria in the DSM-5 and the aforementioned evaluation results. All participants with confirmed ASD (the ASD group) reached the cut-off scores for ASD diagnosis for both evaluation scales. Analytic Approach The sex difference between the HR and TD groups was compared using the χ^2 test. The differences between the HR group and the TD group in social behaviors were determined using the independent samples t-test. The correlations of social behaviors in the HR group with age, DQ, and symptom severity were analyzed using Pearson's rho. Finally, models for early ASD screening were constructed using machine learning methods based on ASD group and TD group, and the HR group (contained 40 children with confirmed ASD and 5 children with not met the criteria of ASD) would be used to verify the effectiveness of models for early ASD screening. $P < 0.05$ indicated that the difference was statistically significant. RESULTS Comparison of the General Conditions Between the HR Group and the TD Group Age (months), adaptive DQ, language DQ, fine motor DQ, and personal-social DQ were significantly different ($P < 0.05$) between the HR group and the TD group, while sex DQ and gross motor DQ were not significantly different ($P > 0.05$) between the two groups. Page number: 4

Score: 0.6262

Text:

LHTbasis of communication and interpersonal relationships with others. Abnormal expression is a prominent manifestation of autism, and it is also one of the criteria for the diagnosis of autism. Doctors can diagnose autism by responding to abnormal facial expressions in children.is a representative disease of generalized developmental disorders. In recent years, the incidence of autism in children has become higher and higher, experiencing a transition from rare diseases to epidemics. At present, research on autism is still in its infancy at home and abroad, and research methods and tools are still developing. The main symptoms of autism include impaired social and interpersonal communication, language retardation, repetitive behavior and sensory dysfunction. It is difficult for autistic patients to correctly recognize faces and explain facial emotions. They have different emotional expressions from ordinary people, and they cannot correctly perceive and understand some basic expressions such as anger (Yan, 2008). At present, the diagnostic methods for autism spectrum disorders include: traditional standard DSM-IV-TR (Segal, 2010) and ICD-10 (Organization W H, 1992), various autism diagnostic assessment scales such as "Childhood Autism Rating Scale (CARS)", "the autism child behavior scale (ABC)" and autism behavior rating scale and questionnaire interviews (Wang and Lu, 2015). Most of these methods rely on doctors' direct observation of the patient's expression, speech and behavior based on their experience. Diagnostic results are easily disturbed by external factors such as hospital level, physician's subjective level, patient's education level, age and so on. Page number: 2

5 Functionizing our semantic search pipeline

Let's put all of the steps from above for semantic search into a function or two so we can repeat the workflow.

```
[43]: def retrieve_relevant_resources(query: str,
                                     embeddings: torch.tensor,
                                     model: SentenceTransformer=embedding_model,
                                     n_resources_to_return: int=5,
                                     print_time: bool=True):
    """ Embeds a query with model and returns top k scores and indices from
    ↪ embeddings. """
    # Embed the query
    query_embedding = model.encode(query, convert_to_tensor=True)
    # Get dot product scores on embeddings
    start_time = timer()
    dot_scores = util.dot_score(query_embedding, embeddings)[0]
    end_time = timer()

    if print_time:
        print(f"[INFO] Time taken to get scores on {len(embeddings)} embeddings:
        ↪ {end_time-start_time:.5f} seconds.")

    scores, indices = torch.topk(input=dot_scores, k=n_resources_to_return)
    return scores, indices

[44]: def print_top_results_and_scores(query: str,
                                       embeddings: torch.tensor,
                                       pages_and_chunks: list[dict]=pages_and_chunks,
                                       n_resources_to_return: int=5):
    """ Takes a query, retrieves most relevant resources and prints them out in
    ↪ descending order.
    Note: Requires pages_and_chunks to be formatted in a specific way (se above
    ↪ for reference). """
    scores, indices = retrieve_relevant_resources(query=query,
    ↪ embeddings=embeddings,
    ↪ n_resources_to_return=n_resources_to_return)

    print(f"Query: {query}\n")
    print("Results:")
    # Loop through zipped together scores and indicies

    for score, index in zip(scores, indices):
        print(f"Score: {score:.4f}")
        # Print relevant sentence chunk (since the scores are in descending order,
        ↪ the most relevant chunk will be first)
        print_wrapped(pages_and_chunks[index]["sentence_chunk"])
```

```

    # Print the page number too so we can reference the textbook further and
    ↪check the results
    print(f"Page number: {pages_and_chunks[index]['page_number']}")
    print(f"PDF name: {pages_and_chunks[index]['pdf_name']}")
    print("\n")

```

6 Now let's test our functions out.

```

[45]: query = "Proposed method for detection of Infants with Autism Spectrum Disorder
    ↪Using Auto-Encoder Feature Representation"
    # Get just the scores and indices of top related results
    scores, indices = retrieve_relevant_resources(query=query,
    ↪embeddings=embeddings)
    scores, indices

```

Batches: 0%| | 0/1 [00:00<?, ?it/s]

[INFO] Time taken to get scores on 503 embeddings: 0.00033 seconds.

```

[45]: (tensor([0.7837, 0.6978, 0.6846, 0.6817, 0.6805]),
    tensor([ 0, 17, 164, 47, 158]))

```

```

[46]: # Print out the texts of the top scores
    print_top_results_and_scores(query=query, embeddings=embeddings)

```

Batches: 0%| | 0/1 [00:00<?, ?it/s]

[INFO] Time taken to get scores on 503 embeddings: 0.00029 seconds.

Query: Proposed method for detection of Infants with Autism Spectrum Disorder
Using Auto-Encoder Feature Representation

Results:

Score: 0.7837

.sensorsbyLetterLetter Deep-Learning-Based Detection of Infants with Autism Spectrum Disorder Using Auto-Encoder Feature RepresentationJung Hyuk Lee 1, Geon Woo Lee 1, Guiyoung Bong 2, Hee Jeong Yoo 2,3 and Hong Kook Kim 1,*1School of Electrical Engineering and Computer Science, Gwangju Institute of Science and Technology, Gwangju 61005, Korea; ljh0412@gist.ac.kr (J. H. L.); geonwoo0801@gist.ac.kr (G. W. L.) 2 Department of Psychiatry, Seoul National University Bundang Hospital, Seongnam-si, Gyeonggi-do 13620, Korea; 20409@snubh.org (G. B.); hjyoo@snu.ac.kr (H. J. Y.)3 Department of Psychiatry, College of Medicine, Seoul National University, Seoul 03980, Korea * Correspondence: hongkook@gist.ac.krReceived: 29 October 2020; Accepted: 24 November 2020; Published: 26 November 2020check for v updatesAbstract: Autism spectrum disorder (ASD) is a developmental disorder with a life-span disability. While diagnostic instruments have been developed and qualified based on the accuracy of the discrimination of children with ASD from typical development

(TD) children, the stability of such procedures can be disrupted by limitations pertaining to time expenses and the subjectivity of clinicians. Consequently, automated diagnostic methods have been developed for acquiring objective measures of autism, and in various fields of research, vocal characteristics have not only been reported as distinctive characteristics by clinicians, but have also shown promising performance in several studies utilizing deep learning models based on the automated discrimination of children with ASD from children with TD. However, difficulties still exist in terms of the characteristics of the data, the complexity of the analysis, and the lack of arranged data caused by the low accessibility for diagnosis and the need to secure anonymity. In order to address these issues, we introduce a pre-trained feature extraction auto-encoder model and a joint optimization scheme, which can achieve robustness for widely distributed and untrained data using a deep-learning-based method for the detection of autism that utilizes various models. By adopting this auto-encoder-based feature extraction and joint optimization in the extended version of the Geneva minimalistic acoustic parameter set (eGeMAPS) speech feature data set, we acquire improved performance in the detection of ASD in infants compared to the raw data set. Keywords: auto-encoder; bidirectional long short-term memory (BLSTM); joint optimization; acoustic feature extraction; autism spectrum disorder. Introduction Autism spectrum disorder (ASD) is a developmental disorder with a high probability of causing difficulties in social interactions with other people [1]. According to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), ASD involves several characteristics such as being con ned to specific interests or behaviors, delayed linguistic development, and poor functionality in terms of communicating or functioning in social situations [2].

Page number: 1

PDF name: LEE.pdf

Score: 0.6978

In addition to ASD detection, this research can be applied to the detection of infants with development delays. Author Contributions: All authors discussed the contents of the manuscript. H. K. K. contributed to the research idea and the framework of this study; G. B. and H. J. Y. provided the database and helped with the discussion; J. H. L. performed the experiments; G. W. L. contributed to the data collection and pre-processing. All authors have read and agreed to the published version of the manuscript. Funding: This work was supported by the Institute of Information & communications Technology Planning & evaluation (IITP) grant funded by the Korea government (MSIT) (No.2019-0-00330, Development of AI Technology for Early Screening of Infant/Child Autism Spectrum Disorders based on Cognition of the Psychological Behavior and Response). Conflicts of Interest: The authors declare no conflict of interest.

Page number: 9

PDF name: LEE.pdf

Score: 0.6846

We used a dataset of 5 children with ASD and 4 TD children older than two years. The accuracy of the proposed method is 96.17% using k-fold cross validation without considering subject-wise hold out, which is a shortcoming of this study. In other words, it has been overfitted to the available data and may fail to correctly classify new samples. So, a thorough examination using an unseen test set on cry features is necessary to evaluate the results. It should be noted that the data from our previous study [41] could not be used in the study presented in this paper due to the differences in data collection procedures. Children with ASD from TD children, are common among all the ASD cases. However, this may not be the case for all the features. For instance, tiptoe walking, which is one of the repetitive behaviors of children with ASD, appears in approximately 25% of these children [42]. Consequently, in the current study, we propose a new cry-based approach for screening children with ASD. Our screening approach makes use of the assumption that all discriminative characteristics of autism may not appear in all ASD children. This assumption is in contrast with the assumption put forward in the ordinary instance-based machine learning methods, which assumes that all instances of a class include all discriminative features needed for classification.

Page number: 3

PDF name: Asd_Cry_patterns.pdf

Score: 0.6817

The 45 children in the HR group were reclassified into two groups after follow-up: 15 children in the N-ASD group who were not meet the criterion of ASD and 40 children in the ASD group. The results showed that the accuracy of Support Vector Machine (SVM) classification was 83.35% for the SF episode. Conclusion: The use of the social behavior indicator of the SFP for a child with HR before 2 years old can effectively predict the clinical diagnosis of the child at the age of 2 years. The screening model constructed using SVM based on the SF episode of the SFP was the best. This also proves that the SFP has certain value in high-risk autism spectrum disorder screening. In addition, because of its convenient, it can provide a self-screening mode for use at home. Trial registration: Chinese Clinical Trial Registry, ChiCTR-OPC-17011995. Keywords: high-risk autism spectrum disorder, the Still-Face Paradigm, social behavior, machine learning, model for early screening. Frontiers in Pediatrics | www.frontiersin.org

Page number: 1

PDF name: Qiu.pdf

Score: 0.6805

Then, after preprocessing, the approach was used to train a classifier, using data for 10 boys with ASD and 10 Typically Developed (TD) boys. The trained classifier was tested on the data of 14 boys and 7 girls with ASD and 14 TD boys and 7 TD girls. The sensitivity, specificity, and precision of the proposed approach for boys were 85.71%, 100%, and 92.85%, respectively. These measures were 71.42%, 100%, and 85.71% for girls, respectively. It was shown that the proposed approach outperforms the common classification methods. Furthermore, it

demonstrated better results than the studies which used voice features for screening ASD. To pilot the practicality of the proposed approach for early autism screening, the trained classifier was tested on 57 participants between 10 to 18 months. These 57 participants consisted of 28 boys and 29 girls and the results were very encouraging for the use of the approach in early ASD screening. Introduction Children with Autism Spectrum Disorder (ASD) are defined by their abnormal or impaired development in social interaction and communication, as well as restricted and repetitive behaviors, interests, or activities [1]. The rapid growth of ASD in the past 20 years has inspired many research efforts toward the diagnosis and rehabilitation of ASD [2-5].

Page number: 1

PDF name: Asd_Cry_patterns.pdf

7 Finally let's set up some query and get the most relevant results.

```
[47]: questions = ['''How does the system architecture in the developed work utilize
ROS for robot-centered systems, and what are the key features of this framework?
↪''',
                  '''How effective is the Video-referenced Infant Rating System for
Autism (VIRSA) in identifying autism spectrum disorder (ASD) risk_
↪in infancy?''',
                  '''What are the key psychometric properties of the_
↪Video-referenced Infant Rating System
for Autism (VIRSA) that were examined in the study, and what were_
↪the findings?''']
```

```
[48]: for question in questions:
print_top_results_and_scores(query=question, embeddings=embeddings)
```

Batches: 0% | 0/1 [00:00<?, ?it/s]

[INFO] Time taken to get scores on 503 embeddings: 0.00089 seconds.

Query: How does the system architecture in the developed work utilize ROS for robot-centered systems, and what are the key features of this framework?

Results:

Score: 0.5763

Fig.1 Node graph architecture of the proposed ROS-based system. The system is composed of two interconnected modules, an artificial reasoning module and a CRI-channel module. The ONO web server has two way of bidirectional communication: a websocket and a standard ROS SubscriberROS Manager /ROS_core /Kinect2_bridge g /Fusion_Node /kinect2_n/hd/ image_color_rect ry Sound /ROS_bridge Controller /tf_transform websocket /Detect_Recognize X /nav_msgs/ Head_Odomets ONO Web Server g /NFOA_Node /known_face n /std_msgs/header/ /ONO_Node Servo g vfoa_msgs Controller /CLNF_node 8 Artificial Reasoning Module

Child-Robot Interaction Modul T Topic Wire Topic Wireless - Local leo & oI Connection Connection 5, ", developed in the server-side application to directly run a ROS node and communicate with standard ROS publishers and subscribers. The CRI is implemented through the open source platform for social robotics (OPSORO),> which is a promising and straightforward system developed for face to face communication composed of a low-cost modular robot called ONO (see Fig.2) and web-based applications [40]. Some of the most important requirements and characteristics that make ONO interesting for this CRI strategy are explained in the following sections. The robot is covered in foam and also fabric to have a more inviting and huggable appearance to the children. The robot has an oversized head to make its facial expressions more prominent and to highlight the importance for communication and emotional interaction. As a consequence of its size and pose, children can interact with the robot at eye height when the robot is placed on a table. The robot ONO has not a predefined identity, as the only element previously conceived is the name.

Page number: 4

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Score: 0.5463

implemented to provide some level of artificial cognition to enable automated robot behavior. In contrast with the aforementioned researches, other works implemented automated face analysis and artificial cognition through robot-mediator and computer vision, which analyzed child's engagement [36, 37], emotions recognition capability [13, 15, 38] and child's intentions [14, 16]. In these works, two different strategies were implemented, where the most common is based on mono-camera approach using an external RGB or RGBd sensor [15, 36, 37] or using on-board RGB cameras mounted on the robotic-platform [13, 16]. Other strategies are based on a highly structured environment composed of an external camera plus an on-board camera [38] or a network of vision sensors attached to a small table [14]. These strategies based on multi-camera methods improve the system's performance, but remain constrained in relation to desired features, such as flexibility, scalability, and modularity. Thus, despite the potential that these techniques have shown, achieving automated child's behavior analysis in a naturalistic way into unstructured clinical-setups with robots that interact accordingly remains a challenge in CRI.

3 System Architecture Overview

The ROS system used in this work is a flexible and scalable open framework for writing modular robot-centered systems. Similar to a computing operating system, ROS manages the interface between robot hardware and software modules and provides common device drivers, data structures and tool-based packages, such as visualization and debugging tools. In addition, ROS uses an interface definition language (IDL) to describe the messages sent between process or nodes, this feature facilitates the multi-language (C++, Python and Lisp) development [39]. The overall system developed here was built using a node graph architecture, taking advantages of the principal ROS design criteria. As with ROS, our system consists of a number of nodes to local video processing together a robot's behavior estimation, distributed around a number of different hosts and connected at runtime in a peer-to-peer topology. The inter-node

connection is implemented as a hand-shaking and occurs in XML-RPC protocol along with a web-socket communication for robot's web-based node (/ONO_node, see Fig. Page number: 3

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Score: 0.5416

1). The node structure is flexible, scalable and can be dynamically modified, i.e., each node can be started and left running along an experimental session or resumed and connected to each other at runtime. In addition, from a general perspective, any robotic platform with web-socket communication can be integrated. The developed system is composed of two interconnected modules as shown in Fig.1: an artificial reasoning module and a CRI-channel module. The module architectures are detailed in the following subsections.3.1 Architecture of Reasoning ModuleIn this module, a distributed architecture for local video processing is implemented. The data of each RGBD sensor in the multi-camera system are processed for two nodes, in which the first is a driver level node and the second 1 node transforms the is a processing node. The driver streaming data of the RGBD sensor into the ROS messages format. The driver addresses the data through a specialized transport provided by plugings to publishes images in a compressed representations while the receptor node only sees sensor_msgs/Image messages. The data processing node executes the face analysis algorithm.

Page number: 3

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Score: 0.5395

This node uses a image_transport subscriber and a ROS packages called CvBridge to turn the data into a image format supported for the typical computer vision algorithms. Later, the same node publishes the head pose and eye gaze direction by means of a ROS navigation message defined as nav_msgs/Odometry. An additional node hosted in the most powerful workstation carries out a data fusion of all navigation messages that were generated in the local processing stage. In addition to the fusion, this node computes the visual focus of attention (VFOA) and publishes this as a std_msgs/Header, in which the time stamp and the target name of the VFOA estimation are registered.3.2 Architecture of CRI-ChannelThe system proposed here has two bidirectional communication channels, a robot-device, and a web-based application to interact with both the child and the therapist. The robot device can interact with the CwASD executing different physical actions, such as facial expression, upper limb poses, and verbal communication. Thus, according to the child's performance, the reasoning module can modify the robot's behavior through automatic gaze shifting, changing the facial expression and providing sound rewards. The client-side application was developed to allow the therapist to control and register all step of the intervention protocol. This interface was also used to supervise and control the robot's behavior and to offer feedback to the therapist about the child's performance along the intervention. This App has two channels of communication for interacting with the reasoning module. The first connection uses a web-

socket protocol and a RosBridge_suite package to support the interpretation of ROS messages, as well as, JSON-based commands in ROS.

Page number: 3

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Score: 0.4940

The second one uses a ROS module!Tools for using the Kinect One (Kinect V2) in ROS, https://github.com/code-iai/iai_kinect2.@ SpringerContent courtesy of Springer Nature, terms of use apply. Rights reserved.

Page number: 3

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Batches: 0%| | 0/1 [00:00<?, ?it/s]

[INFO] Time taken to get scores on 503 embeddings: 0.00024 seconds.

Query: How effective is the Video-referenced Infant Rating System for Autism (VIRSA) in identifying autism spectrum disorder (ASD) risk in infancy?

Results:

Score: 0.8611

HHS Public Access Author manuscript J Child Psychol Psychiatry. Author manuscript; available in PMC 2021 January 01. Published in final edited form as: J Child Psychol Psychiatry. 2020 January ; 61(1): 88-94.doi:10.1111/jcpp.13105. A Video-Based Measure to Identify Autism Risk in Infancy Gregory S. Young, PhDa, John N. Constantino, MDb, Simon Dvorak, BSc, Ashleigh Belding, MPHa, Devon Gangi, PhDa, Alesha Hill, BAa, Monique Hill, MAa, Meghan Miller, PhDa, Chandni Parikh, PhDa, AJ Schwichtenberg, PhDd, Erika Solis, BSa, Sally Ozonoff, PhDa aDepartment of Psychiatry & Behavioral Sciences, MIND Institute, University of California-Davis bDepartment of Psychiatry, Washington University-St. Louis School of Medicine cInformation and Educational Technology, University of California-Davis dDepartment of Human Development & Family Studies, Purdue University Background: Signs of autism are present in the first two years of life, but the average age of diagnosis lags far behind. Instruments that improve detection of autism risk in infancy are needed. This study developed and tested the psychometric properties of a novel video-based approach to detecting ASD in infancy. Methods: A prospective longitudinal study of children at elevated or lower risk for autism spectrum disorder was conducted. Participants were 76 infants with an older sibling with ASD and 37 infants with no known family history of autism. The Video-referenced Infant Rating System for Autism (VIRSA) is a web-based application that presents pairs of videos of parents and infants playing together and requires forced-choice judgments of which video is most similar to the child being rated. Parents rated participants on the VIRSA at 6, 9, 12, and 18 months of age.

Page number: 1

PDF name: Young_Behavior.pdf

Score: 0.8260

The VIRSA app also asked for confirmation of the child's age in order to present videos from a matching age group. Since multiple video exemplars of each scale point (1 to 10) were available, videos were sampled from the pool without replacement. The UC Davis Institutional Review Board approved the study procedures. Parents signed an informed consent form prior to participation. They completed VIRSA ratings when their child was 6-, 9-, 12-, and 18-months-old and again two weeks later to examine test-retest reliability. An automated email invited parents to the online VIRSA app, which could be accessed by computer or mobile device (e.g., smartphones, tablets). VIRSA ratings were always done prior to in-person assessments, which were conducted at 6, 12, 18, 24, and 36 months by examiners unaware of risk group or previous test results. The VIRSA validation sample consisted of 110 infants (73 with an older sibling with ASD, 37 with no known family history of autism), none of whom supplied videos used in instrument development. Twenty-one children in the familial high-risk (HR) group received a diagnosis of ASD, whereas none of the low-risk (LR) children did. ASD diagnoses were made at any age that a child met DSM-5 criteria, based on all information available.

Page number: 4

PDF name: Young_Behavior.pdf

Score: 0.8053

2011) using a parent- report measure (Wetherby, Brosnan-Maddox, Peace & Newton, 2008) identified 32 infants with ASD. This represents significant under-identification, even after accounting for cases with later onset (Barger, Campbell & McDonough, 2013), since current prevalence studies estimate that 170 of 10,000 children have ASD (Baio et al.,2018). The lower sensitivity of early screening measures may be due to the subtlety of initial ASD symptoms and the difficulty of accurately conveying them to parents through written descriptions. Major sources of error in parent questionnaires include comprehension and interpretation problems (Koriat, Goldsmith & Pansky, 2000; Krosnick & Presser, 2010), such as limited understanding of the queried constructs, inadequate knowledge of developmental milestones, and bias due to post-event information (e.g., eventual diagnosis). The current study moves beyond verbal descriptions by employing video examples to reduce subjective interpretations. The use of videos has been shown to dramatically increase clarity in other fields, from music instruction to motor vehicle repair (Arguel & Jamet, 2009). Recently, video was incorporated in ASD screening by Marrus and colleagues (2015), who had parents complete ratings after watching a video of a socially competent toddler, in order to "reduce discrepant interpretations of items by providing informants with a common naturalistic standard for comparison" (p. 1340). Here we describe the development of a new instrument, the Video-referenced Infant Rating System for Autism (VIRSA). It extends previous approaches (Marrus et al.,2015) byJ Child Psychol Psychiatry. Author manuscript; available in PMC 2021 January 01.

Page number: 2

PDF name: Young_Behavior.pdf

Score: 0.8002

Young et al. stratified by familial risk to yield the HR Non-ASD and LR Non-ASD comparison groups. Descriptive statistics are shown in Table 1. Mullen Scales of Early Learning (MSEL; Mullen, 1995) is an assessment of cognitive, motor, and language development for children aged 1 to 68 months. MSEL scores were used to describe the sample. Scores on the Fine Motor and Visual Reception subtests were used to evaluate discriminant validity since these developmental domains are theoretically unrelated to the social-communicative focus of the VIRSA. Autism Diagnostic Observation Schedule, 2nd edition (ADOS-2; Lord et al., 2012) is an observational measure that assesses ASD symptoms through semi-standardized interactions between the clinician and child. It is comprised of five modules appropriate for various ages and language levels; the Toddler module was administered at 18 and 24 months and either module 1 or 2 was used at 36 months, depending upon the child's verbal level. Analyses utilized the overall total algorithm (Social Affect + Restricted Repetitive Behavior or SARRB) score. Scores on the ADOS at 18 months were also used to examine convergent validity with VIRSA ratings at 18 months. Psychometric properties of the VIRSA were examined in several ways. Split-half reliability was analyzed by comparing the first half of the ratings within a given session to the second half.

Page number: 5

PDF name: Young_Behavior.pdf

Score: 0.7900

Infant Behavior and Development, 38, 107-115. [PubMed: 25656952] Gangi DN, Boterberg S, Schwichtenberg AJ, Solis E, Young GS, Iosif A, & Ozonoff S (2019). Use of prospective longitudinal gaze measurements in defining regression Paper presented at the annual meeting of the International Society for Autism Research, Montreal, 5. Grimes DA, & Schulz KF (2002). Uses and abuses of screening tests. Lancet, 359, 881-884. [PubMed: 11897304] Abbreviations: VIRSA Video-referenced Infant Rating System for Autism

Page number: 8

PDF name: Young_Behavior.pdf

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[INFO] Time taken to get scores on 503 embeddings: 0.00022 seconds.

Query: What are the key psychometric properties of the Video-referenced Infant Rating System

for Autism (VIRSA) that were examined in the study, and what were the findings?

Results:

Score: 0.8379

HHS Public Access Author manuscript J Child Psychol Psychiatry. Author manuscript; available in PMC 2021 January 01. Published in final edited form as: J Child Psychol Psychiatry. 2020 January ; 61(1): 88-94. doi:10.1111/jcpp.13105. A Video-Based Measure to Identify Autism Risk in Infancy Gregory S. Young, PhDa, John N. Constantino, MDb, Simon Dvorak, BSc, Ashleigh Belding, MPH a, Devon Gangi, PhDa, Alesha Hill, BAa, Monique Hill, MAa, Meghan Miller, PhDa, Chandni Parikh, PhDa, AJ Schwichtenberg, PhDd, Erika Solis, BSa, Sally Ozonoff, PhDa aDepartment of Psychiatry & Behavioral Sciences, MIND Institute, University of California-Davis bDepartment of Psychiatry, Washington University-St. Louis School of Medicine cInformation and Educational Technology, University of California-Davis dDepartment of Human Development & Family Studies, Purdue University Background: Signs of autism are present in the first two years of life, but the average age of diagnosis lags far behind. Instruments that improve detection of autism risk in infancy are needed. This study developed and tested the psychometric properties of a novel video-based approach to detecting ASD in infancy. Methods: A prospective longitudinal study of children at elevated or lower risk for autism spectrum disorder was conducted. Participants were 76 infants with an older sibling with ASD and 37 infants with no known family history of autism. The Video-referenced Infant Rating System for Autism (VIRSA) is a web-based application that presents pairs of videos of parents and infants playing together and requires forced-choice judgments of which video is most similar to the child being rated. Parents rated participants on the VIRSA at 6, 9, 12, and 18 months of age.

Page number: 1

PDF name: Young_Behavior.pdf

Score: 0.7901

Infant Behavior and Development, 38, 107-115. [PubMed: 25656952] Gangi DN, Boterberg S, Schwichtenberg AJ, Solis E, Young GS, Iosif A, & Ozonoff S (2019). Use of prospective longitudinal gaze measurements in defining regression Paper presented at the annual meeting of the International Society for Autism Research, Montreal, 5. Grimes DA, & Schulz KF (2002). Uses and abuses of screening tests. Lancet, 359, 881-884. [PubMed: 11897304] Abbreviations: VIRSA Video-referenced Infant Rating System for Autism

Page number: 8

PDF name: Young_Behavior.pdf

Score: 0.7734

Young et al. stratified by familial risk to yield the HR Non-ASD and LR Non-ASD comparison groups. Descriptive statistics are shown in Table 1. Mullen Scales of Early Learning (MSEL; Mullen, 1995) is an assessment of cognitive, motor, and language development for children aged 1 to 68 months. MSEL scores were used to describe the sample. Scores on the Fine Motor and Visual Reception subtests were used to evaluate discriminant validity since these developmental domains are theoretically unrelated to the social-communicative focus of the VIRSA. Autism Diagnostic Observation Schedule, 2nd edition (ADOS-2; Lord et al., 2012) is an

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Page number: 5

PDF name: Young_Behavior.pdf

Score: 0.7705

The VIRSA app also asked for confirmation of the child's age in order to present videos from a matching age group. Since multiple video exemplars of each scale point (1 to 10) were available, videos were sampled from the pool without replacement. The UC Davis Institutional Review Board approved the study procedures. Parents signed an informed consent form prior to participation. They completed VIRSA ratings when their child was 6-, 9-, 12-, and 18-months-old and again two weeks later to examine test-retest reliability. An automated email invited parents to the online VIRSA app, which could be accessed by computer or mobile device (e.g., smartphones, tablets). VIRSA ratings were always done prior to in-person assessments, which were conducted at 6, 12, 18, 24, and 36 months by examiners unaware of risk group or previous test results. The VIRSA validation sample consisted of 110 infants (73 with an older sibling with ASD, 37 with no known family history of autism), none of whom supplied videos used in instrument development. Twenty-one children in the familial high-risk (HR) group received a diagnosis of ASD, whereas none of the low-risk (LR) children did. ASD diagnoses were made at any age that a child met DSM-5 criteria, based on all information available.

Page number: 4

PDF name: Young_Behavior.pdf

Score: 0.7690

2011) using a parent- report measure (Wetherby, Brosnan-Maddox, Peace & Newton, 2008) identified 32 infants with ASD. This represents significant under-identification, even after accounting for cases with later onset (Barger, Campbell & McDonough, 2013), since current prevalence studies estimate that 170 of 10,000 children have ASD (Baio et al., 2018). The lower sensitivity of early screening measures may be due to the subtlety of initial ASD symptoms and the difficulty of accurately conveying them to parents through written descriptions. Major sources of error in parent questionnaires include comprehension and interpretation problems (Koriat, Goldsmith & Pansky, 2000; Krosnick & Presser, 2010), such as limited understanding of the queried constructs, inadequate knowledge of developmental milestones, and bias due to post-event information

(e.g., eventual diagnosis). The current study moves beyond verbal descriptions by employing video examples to reduce subjective interpretations. The use of videos has been shown to dramatically increase clarity in other fields, from music instruction to motor vehicle repair (Arguel & Jamet, 2009). Recently, video was incorporated in ASD screening by Marrus and colleagues (2015), who had parents complete ratings after watching a video of a socially competent toddler, in order to "reduce discrepant interpretations of items by providing informants with a common naturalistic standard for comparison" (p. 1340). Here we describe the development of a new instrument, the Video-referenced Infant Rating System for Autism (VIRSA). It extends previous approaches (Marrus et al., 2015) by J Child Psychol Psychiatry. Author manuscript; available in PMC 2021 January 01. Page number: 2
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7.1 Great!!

7.2 Now you can choose your desired LLM and use it to generate your text.

7.3 Also try the notebook on your own set of PDFs and see how it works!

[]: